



SWIVELS

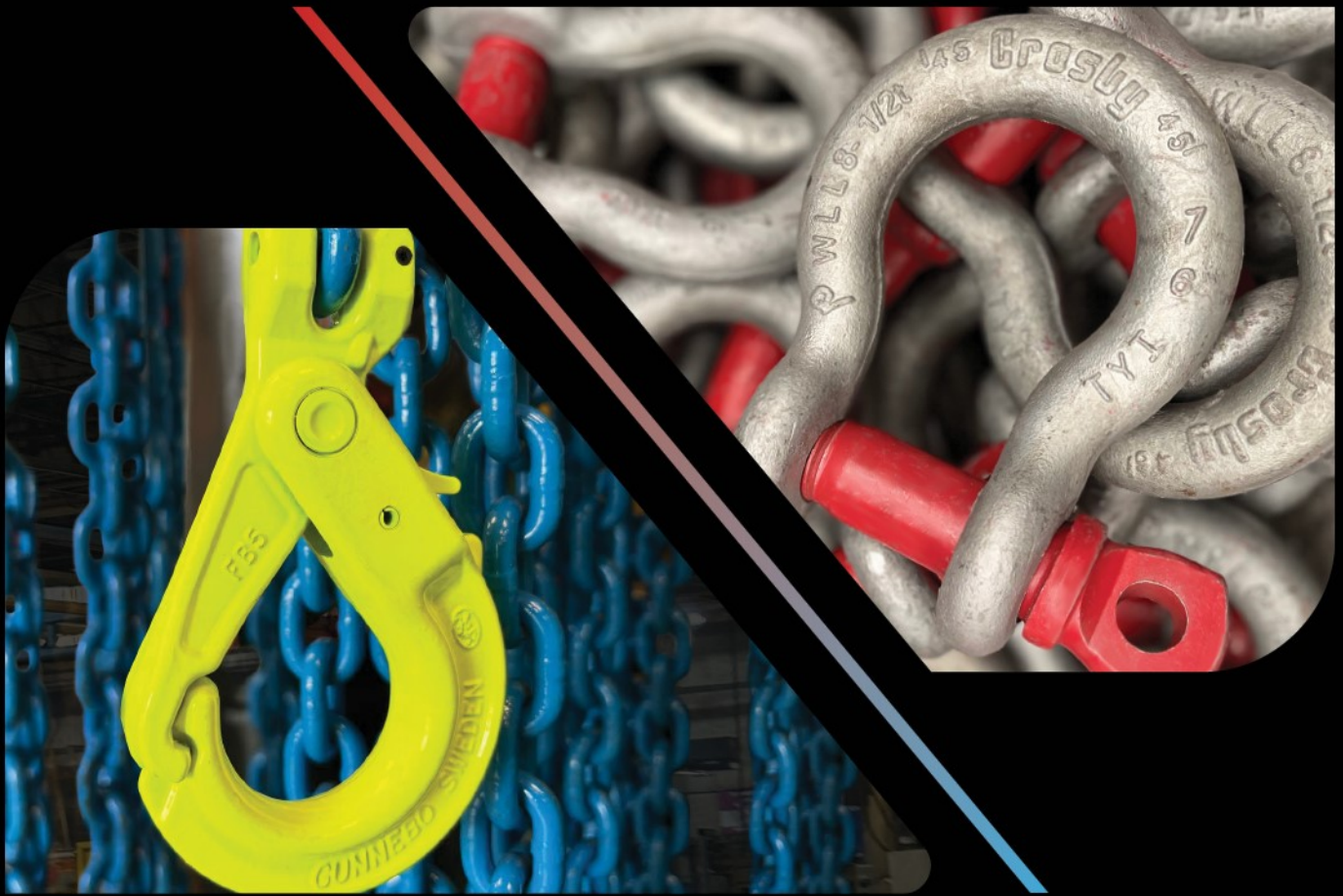
Crosby®

MILLER®

IMPORTED

CoreLifting.com





Crosby®



GUNNEBO®
Industries

PRODUCT CATALOG

with application & warning information

Imperial

Crosby® Forged Swivels

- 402 and 403 forged swivels are positioning devices and are not intended to rotate under load.
- Hot-dip galvanized.
- Quenched & Tempered.
- Crosby products meet or exceed all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, Crosby products meet other critical performance requirements, including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- G-402 swivels meet the performance requirements of Federal Specification RR-C-271G, Type VII, Class 2, except for those provisions required of the contractor.
- G-403 swivels meet the performance requirements of Federal Specification RR-C-271G, Type VII, Class 3, except for those provisions required of the contractor.



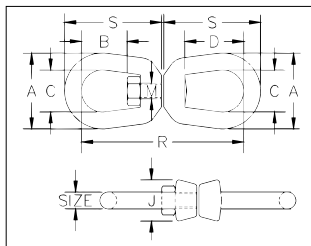
4



1/4" - 1 1/4" size



1 1/2" size



G-402 Regular Swivels

Size (in)	Stock No.	Working Load Limit (lb)*	Weight Each (lb)	Dimensions (in)							
				A	B	C	D	J	M	R	S
5/16	1016037	1250	.39	1.63	.81	1.00	1.25	.81	.38	3.56	2.06
3/8	1016055	2250	.71	2.00	.94	1.25	1.50	1.00	.50	4.31	2.50
1/2	1016073	3600	1.32	2.50	1.31	1.50	2.00	1.31	.63	5.44	3.19
5/8	1016091	5200	2.49	3.00	1.56	1.75	2.38	1.50	.75	6.56	3.88
3/4	1016117	7200	4.02	3.50	1.75	2.00	2.63	1.88	.88	7.19	4.31
7/8	1016135	10000	6.25	4.00	2.06	2.25	3.06	2.13	1.00	8.38	5.00
1	1016153	12500	8.95	4.50	2.31	2.50	3.50	2.38	1.13	9.63	5.75
1-1/4	1016199	18000	16.37	5.63	2.69	3.13	3.69	3.00	1.50	11.44	6.75
1-1/2+	1016215	45200	45.79	7.09	3.88	4.09	3.88	3.75	2.25	16.69	9.91

5:1 Design Factor.



G-403 Jaw End Swivels

Size (in)	Stock No.	Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)												
				A	B	C	G	J	K	L	M	N	P	R	U	V
5/16	1016411	1250	.34	1.63	.81	1.00	.81	.81	.50	1.13	.38	.88	.31	2.94	2.06	1.81
3/8	1016439	2250	.66	2.00	.94	1.25	1.00	1.00	.63	1.41	.50	1.06	.38	3.63	2.50	2.25
1/2	1016457	3600	1.34	2.50	1.31	1.50	1.31	1.31	.75	1.75	.63	1.31	.50	4.50	3.19	2.88
5/8	1016475	5200	2.48	3.00	1.56	1.75	1.63	1.50	.94	2.06	.75	1.50	.63	5.31	3.88	3.44
3/4	1016493	7200	3.88	3.50	1.75	2.00	1.88	1.88	1.13	2.53	.88	1.75	.75	6.06	4.31	4.00
7/8	1016518	10000	5.87	4.00	2.06	2.25	2.13	2.13	1.34	2.79	1.00	2.06	.88	7.00	5.00	4.53
1	1016536	12500	9.84	4.50	2.31	2.50	2.63	2.38	1.75	3.72	1.13	2.81	1.13	8.56	5.75	5.94
1-1/4	1016572	18000	15.75	5.69	2.69	3.13	3.13	3.00	2.06	4.31	1.63	2.81	1.38	9.75	7.06	6.38
1-1/2	1016590	45200	54.75	7.00	3.88	4.00	5.63	4.00	2.88	6.00	2.25	4.44	2.25	14.25	10.00	10.84

5:1 Design Factor.

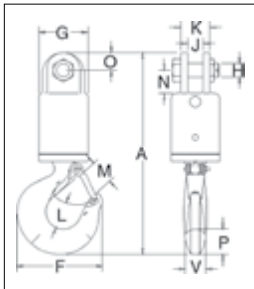


Crosby® Tapered Roller Bearing Swivels

- Equipped with tapered roller thrust bearing.
- Suitable for frequent rotation under load.
- All swivels individually proof tested to 2 times the Working Load Limit with labeled documentation.
- All hooks furnished with latches assembled.
- All jaws complete with bolts, nuts, and cotter pins.
- Pressure lube fitting provided.
- NOT TO BE USED ON DEMOLITION (WRECKING) BALLS.
- Other types and capacities up to 1,250t available to meet your requirements. Visit thecrosbygroup.com/engineeredolutions for more information.
- IMPORTANT - Crosby swivels should only be used with the recommended wire rope. Contact the wire rope manufacturer for the proper wire rope to be used with Crosby swivels.



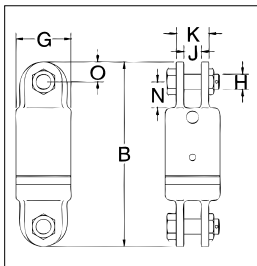
S-1 Jaw & Hook



Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)											
					A	F	G	H	J	K	L	M	N	O	P	V
3-S-1	297011	3	1/2	9.81	11.44	4.84	2.75	.75	.88	1.62	1.53	1.41	1.31	1.00	1.44	1.12
5-S-1	297217	5	5/8	15.51	13.34	6.28	3.00	.88	1.00	2.25	1.94	1.69	1.62	1.12	1.81	1.44
8-S-1	297413	8.5	3/4	29.42	16.45	7.54	4.00	1.00	1.56	2.81	2.46	2.22	2.12	1.38	2.25	1.62
10-S-1	297618	10	7/8	46.75	19.75	8.34	4.50	1.50	1.75	3.38	2.59	2.41	3.50	1.75	2.59	1.94
15-S-1	297814	15	1	73.75	22.24	10.34	5.00	1.50	1.75	3.38	2.81	3.19	3.50	1.75	3.00	2.38
25-S-1	298118	25	-	140.00	26.78	13.62	6.00	2.00	2.00	4.62	3.44	3.62	3.69	2.38	3.66	3.00
35-S-1	298216	35	-	220.00	29.94	14.06	6.50	2.00	2.00	4.62	3.88	3.75	3.69	2.38	4.56	3.19
45-S-1	298314	45	-	251.00	35.06	15.44	7.00	2.25	2.50	5.00	4.75	4.25	4.00	3.00	5.06	3.25

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

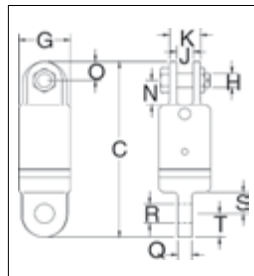
S-2 Jaw & Jaw



Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)						
					B	G	H	J	K	N	O
3-S-2	297020	3	1/2	9.63	9.28	2.75	.75	.88	1.62	1.31	1.00
5-S-2	297226	5	5/8	13.69	10.31	3.00	.88	1.00	2.25	1.62	1.12
8-S-2	297422	8.5	3/4	26.16	12.62	4.00	1.00	1.56	2.81	2.12	1.38
10-S-2	297627	10	7/8	45.75	16.75	4.50	1.50	1.75	3.38	3.50	1.75
15-S-2	297823	15	1	62.75	17.12	5.00	1.50	1.75	3.38	3.50	1.75
25-S-2	298127	25	-	140.00	20.75	6.00	2.00	2.00	4.62	3.69	2.38
35-S-2	298225	35	-	155.00	20.75	6.50	2.00	2.00	4.62	3.69	2.38
45-S-2	298323	45	-	235.00	25.25	7.00	2.25	2.50	5.00	4.00	3.00

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

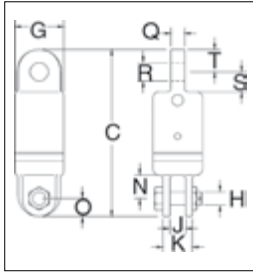
S-3 Jaw & Eye



Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)									
					C	G	H	J	K	N	O	Q	R	T
3-S-3	297039	3	1/2	9.12	9.34	2.75	.75	.88	1.62	1.31	1.00	.75	1.03	1.12
5-S-3	297235	5	5/8	13.50	10.06	3.00	.88	1.00	2.25	1.62	1.12	1.00	1.28	1.25
8-S-3	297431	8.5	3/4	24.90	12.25	4.00	1.00	1.56	2.81	2.12	1.38	1.25	1.41	1.62
10-S-3	297636	10	7/8	43.50	16.12	4.50	1.50	1.75	3.38	3.50	1.75	1.69	1.69	2.75
15-S-3	297832	15	1	61.00	16.75	5.00	1.50	1.75	3.38	3.50	1.75	1.94	2.03	2.75
25-S-3	298136	25	-	135.00	21.50	6.00	2.00	2.00	4.62	3.69	2.38	2.25	2.31	3.88
35-S-3	298234	35	-	150.00	21.50	6.50	2.00	2.00	4.62	3.69	2.38	2.25	2.31	3.88
45-S-3	298332	45	-	225.00	25.88	7.00	2.25	2.50	5.00	4.00	3.00	2.50	2.53	4.00

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

S-4 Eye & Jaw

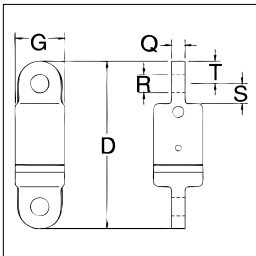


Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)										
					C	G	H	J	K	N	O	Q	R	S	T
3-S-4	297048	3	1/2	9.00	9.34	2.75	.75	.88	1.62	1.31	1.00	.75	1.03	1.12	1.25
5-S-4	297244	5	5/8	12.33	10.06	3.00	.88	1.00	2.25	1.62	1.12	1.00	1.28	1.25	1.25
8-S-4	297440	8.5	3/4	29.00	12.25	4.00	1.00	1.56	2.81	2.12	1.38	1.25	1.41	1.62	1.50
10-S-4	297645	10	7/8	44.00	16.12	4.50	1.50	1.75	3.38	3.50	1.75	1.69	1.69	2.75	1.88
15-S-4	297841	15	1	61.00	16.75	5.00	1.50	1.75	3.38	3.50	1.75	1.94	2.03	2.75	2.12
25-S-4	298145	25	-	135.00	21.50	6.00	2.00	2.00	4.62	3.69	2.38	2.25	2.31	3.88	2.38
35-S-4	298243	35	-	150.00	21.50	6.50	2.00	2.00	4.62	3.69	2.38	2.25	2.31	3.88	2.38
45-S-4	298341	45	-	225.00	25.88	7.00	2.25	2.50	5.00	4.00	3.00	2.50	2.53	4.00	3.00

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

4

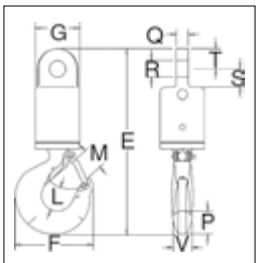
S-5 Eye & Eye



Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)					
					D	G	Q	R	S	T
3-S-5	297057	3	1/2	8.50	9.41	2.75	.75	1.03	1.12	1.25
5-S-5	297253	5	5/8	11.30	9.81	3.00	1.00	1.28	1.25	1.25
8-S-5	297459	8.5	3/4	29.25	11.88	4.00	1.25	1.41	1.62	1.50
10-S-5	297654	10	7/8	42.00	15.50	4.50	1.69	1.69	2.75	1.88
15-S-5	297850	15	1	49.00	16.38	5.00	1.94	2.03	2.75	2.12
25-S-5	298154	25	-	130.00	22.25	6.00	2.25	2.31	3.88	2.38
35-S-5	298252	35	-	145.00	22.25	6.50	2.25	2.31	3.88	2.38
45-S-5	298350	45	-	215.00	26.50	7.00	2.50	2.53	4.00	3.00

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

S-6 Eye & Hook



Swivel No.	Stock No.	Working Load Limit (t)*	Wire Rope Size (in)	Weight Each (lb)	Dimensions (in)										
					E	F	G	L	M	P	Q	R	S	T	V
3-S-6	297066	3	1/2	9.32	11.50	4.84	2.75	1.53	1.41	1.44	.75	1.03	1.12	1.25	1.12
5-S-6	297262	5	5/8	14.24	13.09	6.28	3.00	1.94	1.69	1.81	1.00	1.28	1.25	1.25	1.44
8-S-6	297468	8.5	3/4	32.00	16.07	7.54	4.00	2.46	2.22	2.25	1.25	1.41	1.62	1.50	1.62
10-S-6	297663	10	7/8	45.50	19.12	8.34	4.50	2.59	2.41	2.59	1.69	1.69	2.75	1.88	1.94
15-S-6	297869	15	1	63.00	21.24	10.34	5.00	2.81	3.19	3.00	1.94	2.03	2.75	2.12	2.38
25-S-6	298163	25	-	135.00	27.53	13.62	6.00	3.44	3.62	3.66	2.25	2.31	3.88	2.38	3.00
35-S-6	298261	35	-	215.00	30.69	14.06	6.50	3.88	3.75	4.56	2.25	2.31	3.88	2.38	3.19
45-S-6	298369	45	-	270.00	35.69	15.44	7.00	4.75	4.25	5.06	2.50	2.53	4.00	3.00	3.25

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

Crosby Tapered Roller Bearing Swivels Service Parts

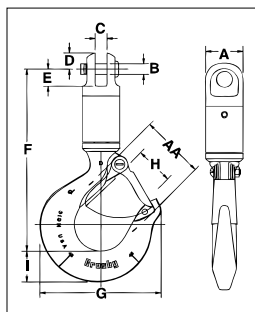
Working Load Limit (t)	Replacement Bolt Kit Stock No.	Replacement Latch Kit Stock No.
3	1411118	1096515
5	1411127	1096562
8.5	1411136	1096609
10	1411145	1096657
15	1411145	1096704
25	134893	1090161
35	134893	1090189
45	134900	1090189

Crosby® Angular Contact Bearing Swivels

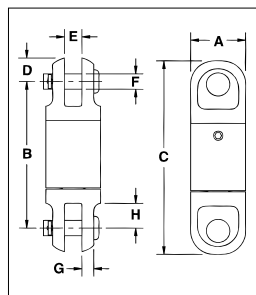


- Designed for high rotation speed, lower torque required to initiate rotation.
- Angular contact bearings maximize efficiency, reliability, and service life of swivel and extend the life of the wire rope.
- Entire swivel is zinc plated to resist corrosion.
- Individually Proof Tested to 2 times the Working Load Limit with certification.
- Hook models utilize genuine Crosby hooks which are forged alloy steel, quenched and tempered, and contain patented QUIC-CHECK® markings.
- Each swivel 8.5 short Tons and larger is furnished with a pressure lubrication fitting.
- For swivels larger than those listed, visit thecrosbygroup.com/engineeredolutions for more information.

AS-1 Jaw & Hook



AS-2 Jaw & Jaw



AS-1 Jaw & Hook

Load Rated

QUIC-CHECK®

CE

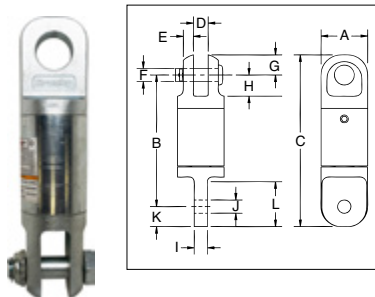
AS-1 JAW & HOOK				Dimensions (in)									Deformation Indicator AA	Replacement Latch Kit Stock No.	Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	H	I			
.40	1/8	1016001	.7	.88	.25	.25	.38	.41	4.32	2.86	.93	.73	1.50	1096325	2018360
.68	1/4	1016010	1.5	1.31	.38	.31	.44	.56	5.44	3.16	.97	.84	1.50	1096374	2018361
1.35	3/8	1016025	2.3	1.63	.50	.53	.69	.78	6.35	4.00	1.16	1.14	1.50	1096374	2018362
2.70	1/2	1016026	6.5	2.00	.75	.75	.94	1.19	8.69	4.84	1.41	1.44	2.50	1096374	2018363
4.50	5/8	1016040	12.9	2.50	.88	1.00	1.13	1.53	10.71	6.28	1.69	1.82	3.00	1096562	2018364
7.65	3/4	1016045	26.4	3.00	1.19	1.56	1.34	2.09	13.65	8.34	2.41	2.60	4.00	1096657	2018365
9.00	7/8	1016056	53.0	4.00	1.50	1.75	1.75	3.50	17.95	10.34	3.19	3.00	5.00	1096704	2018367
13.5	1	1016064	53.0	4.00	1.50	1.75	1.75	3.50	17.95	10.34	3.19	3.00	5.00	1096704	2018367
22.5	1-1/4	1016075	97.0	5.00	2.00	2.00	2.38	3.69	20.88	13.62	3.25	3.62	6.50	1090161	2018368
31.5	1-1/2	1016082	140.0	5.00	2.00	2.00	2.38	3.69	24.00	14.06	3.00	4.56	7.00	1090189	2018368

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

AS-2 Jaw & Jaw

AS-2 JAW & JAW				Dimensions (in)								Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	H	
.40	1/8	1016103	.4	.88	2.38	3.13	.38	.25	.25	.19	.41	2018360
.68	1/4	1016114	.9	1.31	3.56	4.44	.44	.31	.38	.22	.56	2018361
1.35	3/8	1016122	2.0	1.63	4.06	5.44	.69	.50	.50	.28	.78	2018362
2.70	1/2	1016131	4.9	2.00	6.25	8.13	.94	.75	.75	.38	1.19	2018363
4.50	5/8	1016139	9.6	2.50	7.75	10.63	1.13	1.00	.88	.53	1.53	2018364
7.65	3/4	1016148	15.8	3.00	9.63	12.31	1.34	1.56	1.19	.56	2.09	2018365
9.00	7/8	1016157	40.0	4.00	14.00	17.50	1.75	1.75	1.50	.81	3.50	2018367
13.5	1	1016166	40.0	4.00	14.00	17.50	1.75	1.75	1.50	.81	3.50	2018367
22.5	1-1/4	1016175	78.0	5.00	15.94	20.69	2.38	2.00	2.00	1.13	3.69	2018368
31.5	1-1/2	1016184	78.0	5.00	15.94	20.69	2.38	2.00	2.00	1.13	3.69	2018368

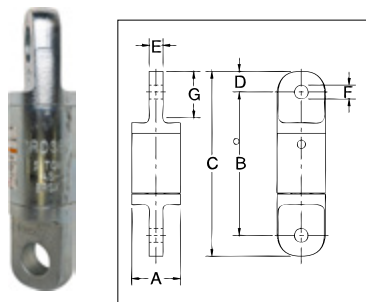
5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.



AS-3 Jaw & Eye

AS-3 JAW & EYE				Dimensions (in)												Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	H	I	J	K	L	
.40	1/8	1016205	.3	.88	2.50	3.25	.25	.19	.25	.38	.41	.25	.25	.38	.84	2018360
.68	1/4	1016216	.9	1.31	3.69	4.56	.31	.22	.38	.44	.56	.31	.38	.44	.88	2018361
1.35	3/8	1016224	1.9	1.63	4.19	5.44	.50	.28	.50	.69	.78	.50	.66	.63	1.38	2018362
2.70	1/2	1016232	4.6	2.00	6.19	8.13	.75	.38	.75	.94	1.19	.75	.91	1.00	2.00	2018363
4.50	5/8	1016243	9.1	2.50	7.88	10.19	1.00	.53	.88	1.13	1.50	1.00	1.25	1.19	2.63	2018364
7.65	3/4	1016250	15.6	3.00	9.50	12.25	1.56	.56	1.25	1.34	2.09	1.25	1.41	1.50	3.13	2018365
9.00	7/8	1016259	39.0	4.00	13.75	17.31	1.75	.81	1.50	1.75	3.50	1.72	1.63	1.81	4.69	2018367
13.5	1	1016268	40.0	4.00	13.44	17.31	1.75	.81	1.50	1.75	3.50	2.00	2.00	2.13	4.69	2018367
22.5	1-1/4	1016277	78.0	5.00	16.00	20.75	2.00	1.13	2.00	2.38	3.69	2.25	2.31	2.38	5.25	2018368
31.5	1-1/2	1016286	78.0	5.00	16.00	20.75	2.00	1.13	2.00	2.38	3.69	2.25	2.31	2.38	5.2	2018368

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

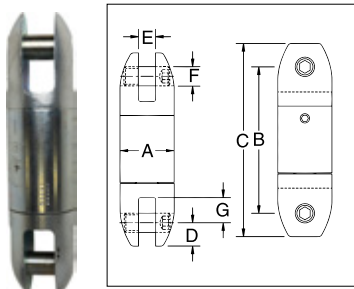


AS-5 Eye & Eye

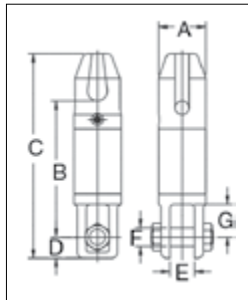
AS-5 EYE & EYE				Dimensions (in)						
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G
.40	1/8	1016409	.3	.88	2.63	3.38	.38	.25	.25	.81
.68	1/4	1016418	.9	1.31	3.75	4.63	.44	.31	.38	.88
1.35	3/8	1016427	1.8	1.63	4.31	5.56	.63	.50	.66	1.34
2.70	1/2	1016436	4.3	2.00	6.13	8.13	1.00	.75	.91	2.00
4.50	5/8	1016445	8.6	2.50	7.75	10.63	1.19	1.00	1.25	2.63
7.65	3/4	1016454	15.4	3.00	9.31	12.31	1.50	1.25	1.41	3.13
9.00	7/8	1016463	37.0	4.00	13.88	17.50	1.81	1.72	1.63	4.69
13.5	1	1016472	39.0	4.00	13.25	17.50	2.13	2.00	2.13	4.69
22.5	1-1/4	1016481	72.0	5.00	16.00	20.75	2.38	2.25	2.31	5.25
31.5	1-1/2	1016490	72.0	5.00	16.00	20.75	2.38	2.25	2.31	5.25

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

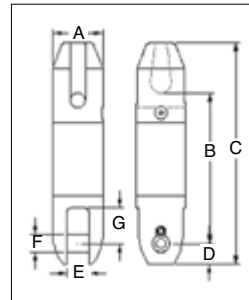
AS-7 Bullet Style Jaw & Jaw



AS-11 Thimble & Jaw



AS-14 Thimble & Bullet



AS-7 Bullet Style Jaw & Jaw

AS-7 BULLET STYLE JAW & JAW				Dimensions (in)							Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	
.40	1/8	1016604	.4	.88	2.38	3.13	.38	.25	.31	.40	2003936
.68	1/4	1016611	1.1	1.31	3.56	4.44	.44	.31	.38	.56	2003904
1.35	3/8	1016622	1.8	1.63	4.06	5.19	.56	.50	.44	.81	2003905
2.70	1/2	1016631	3.8	2.00	5.44	7.06	.81	.75	.63	.94	2003906
4.50	5/8	1016640	8.0	2.50	7.75	10.06	1.13	1.00	.88	1.56	2003907
7.65	3/4	1016649	14.5	3.00	9.88	12.38	1.25	1.31	1.00	2.13	2003943
9.00	7/8	1016652	40.0	4.00	13.13	16.75	1.75	1.75	1.50	3.25	2003959
13.5	1	1016658	40.0	4.00	13.13	16.75	1.75	1.75	1.50	3.25	2003959
22.5	1-1/4	1016662	84.0	5.00	15.94	20.75	2.38	2.00	2.00	3.69	2003935
31.5	1-1/2	1016667	84.0	5.00	15.94	20.75	2.38	2.00	2.00	3.69	2003935

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

AS-11 Thimble & Jaw

AS-11 THIMBLE & JAW				Dimensions (in)							Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	
7.65	3/4	1017020	18.0	3.00	8.66	13.00	1.34	1.56	1.19	2.09	2018365
13.5	1	1017029	42.0	4.00	11.66	17.53	1.75	1.78	1.50	3.50	2018367

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.

AS-14 Thimble & Bullet

AS-14 THIMBLE & BULLET				Dimensions (in)							Replacement Bolt Kit Stock No.
Working Load Limit (t)*	Wire Rope Size (in)	Stock No.	Weight Each (lb)	A	B	C	D	E	F	G	
7.7	3/4	1017255	20.0	3.00	9.00	13.25	1.25	1.31	1.00	2.13	2003943
13.6	1	1017258	40.0	4.00	11.50	17.38	1.75	1.75	1.50	3.25	2003959

5:1 Design Factor. Individually Proof Tested to 2 times the Working Load Limit.



GENERAL CATALOG

LIFTING PRODUCTS

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LIFTING PRODUCTS

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GLOSSARY

Important Terms Defined:

DESIGN (SAFETY) FACTOR: The Ultimate Load divided by the Working Load Limit expressed as a ratio, e.g., 5:1. The Design Safety Factor (or Design Factor) denotes a product's theoretical reserve capability and is not the load rating of the product. See Working Load Limit below.

MEAN BREAKING LOAD (MBL): Similar to Ultimate Load, also the average load at which the product no longer supports the load or fails, used more typically with respect to chain and chain-related hardware.

PROOF LOAD: The average load to which a product may be subjected before physical deformation occurs. This is also the average force applied during a Proof Load Test. Miller products in this catalog have a maximum Proof Load of twice the Working Load Limit, except where otherwise indicated.

PROOF LOAD TEST: A load test of the product conducted at the specified Proof Load for the purpose of detecting material or manufacturing defects.

SHOCK LOADING: A dynamic increase in loading on the product caused by the sudden take up of slack, shifting or jerking of the load, or impact on the load or the product resulting in a significant increase beyond the static load.

SHORT TON (TON) - The unit of mass on which most Miller Lifting Products' capacities, load limits, etc. are based. Commonly used throughout the U.S. and Canada, the short ton equals 2,000 lbs and differs from the "metric ton" (SI) which equals 2,205 lbs and the "long ton" (UK tonne) which equals 2,240 pounds.

METRIC TON (MT) - The SI unit of mass equal to 1000 kilograms or 2,205 lbs. Some Miller Lifting Products capacities, load limits, etc. are expressed in metric tons.

LONG TON (TONNE) - The traditional UK unit of mass equal to 2,240 lbs or 1016 kg. No Miller Lifting Products capacities, load limits, etc. are expressed in long tons. Rarely used outside the UK, the long ton ceased to be legal for trade in 1985.

STATIC LOAD: The load resulting from a constant applied force.

WORKING LOAD: The maximum load which the product is authorized to support in a specific application, equal to or less than the Working Load Limit (WLL).

WORKING LOAD LIMIT (WLL): The maximum static load the product is intended to support in general service with the load being applied uniformly in direct axial tension relative to the centerline of the product. Working Load Limits do not apply to shock loading, hook tip loading, side loading, bending, torsional or other non-axial loading. The Working Load Limit reflects Miller's engineering evaluation and should never be exceeded regardless of the size or strength of the wire rope, synthetic rope, chain, etc. This term is used interchangeably with the following terms:

- Safe Working Load (SWL)
- Safe Working Limit
- Rated Load Value
- Maximum Load Limit
- Resultant Working Load

ULTIMATE LOAD: The average load at which the product fails or no longer supports the load.

YIELD POINT: The point between the Proof Load and the Ultimate Load at which permanent deformation occurs. Note that this deformation may not be visible.

WARNINGS AND PRECAUTIONS

All ratings shown in any Miller literature are based on the products being in a new and unused condition. When selecting a product, consideration must be given to the greatest load the product will see, therefore, shock loading must be considered in the overall system design.

All products manufactured by Miller Products are sold with the express understanding that the customer is thoroughly familiar with the safe and proper use and application of the product.

In selecting a Miller product, such factors as extreme heat or cold, excess humidity, moisture, contaminated air, etc. must be considered since these may adversely affect longevity, performance and *Working Load Limit*.

Do not immerse any standard Miller product in water. Contact the factory for assistance in selecting the special product designed to meet fresh and saltwater applications.

Lift only those loads for which the Miller product is designed. Federal crane regulations prohibit the transport of personnel on any load or wire rope attachment (OSHA 1910, 180-h-3-v).

Miller products are generally designed for applications of tension or straight pull. Side loading must be avoided since it exerts additional force which the product is not designed to accommodate.

HOOK LATCHES - All Miller products that utilize hooks are equipped at the factory with hook latches. The only function of a hook latch is to retain loose slings or devices under slack conditions, i.e. no load. They are not intended to be anti-fouling devices, so caution should be used to prevent the latch from supporting any of the load. Routine inspection of the latches must be made to ensure their proper operating conditions. Hook applications might require other hook latches. Should you have any questions regarding hook latches, contact the factory.

WHENEVER WEDGE SOCKETS ARE UTILIZED THESE TWO PRECAUTIONS MUST BE TAKEN:

1. When installing wire rope always pre-load the wedge with the wire rope in place. Then attach the end of the rope to the main line with the first clip to be located directly above the socket. It is imperative that the clips and clamps be installed precisely as specified by the manufacturers and in the number and sizes approved by the fitting manufacturers.
2. Make allowance for the crimping effect common with all types of wedge sockets. Experience shows the *Maximum Load Limit* on a line will be reduced by 20% or more with this type of fitting.

NEVER WELD ON ANY MILLER PRODUCT. Should any modification or repairs be required on any Miller product, contact the factory for information.

INSPECT AND MAINTAIN REGULARLY. All Miller products are mechanical components and are subject to wear. Worn components do not have the same *Working Load Limit* as do new components. The total responsibility for the inspection, maintenance and continued use is entirely up to the purchaser/user. Remember, visual inspection may not be sufficient and examination methods such as X-ray, ultrasonic testing, magnetic particle inspection, dielectric resistance and others, might be required to establish the present integrity of the product. Check to see that your equipment is being inspected and tested in accordance with all applicable governmental rules and regulations. Should any Miller products become worn and in need of repair, the responsibility for the actual repair work will be borne solely by the party making such repairs. It is recommended that the factory be contacted should there be any questions whatsoever relating to a repair. See Maintenance and Inspection.

INSULATING LINKS – ELECTRICAL PERFORMANCE OF INSULATING LINKS DEGRADES WHEN LINKS ARE NOT KEPT FREE OF SURFACE DIRT AND CONTAMINATION. INSULATING LINK VOLTAGE RATINGS ASSUME CLEAN AND DRY LINKS AND A DRY ENVIRONMENT. In the case of ELECTRICAL POWER LINE SAFETY use of an insulating link alone in the absence of other required measures is NOT ACCEPTABLE. OSHA power line safety requirements involving the use of cranes and derricks can be found in US federal regulation 29 CFR Part 1926, latest edition.

REMEMBER: External factors will affect the longevity of the product. There is no defined period for the useful life of any Miller product. It is the user's responsibility to maintain and check the product. Even after a short period of use circumstances may require the product to be withdrawn from service. Should any questions arise during the inspection of any Miller product that relate to its intended application or need for repair, promptly remove the product from service

PROOF LOAD – The recommended proof load on all items in this catalog is 2 times the *Working Load Limit* unless otherwise indicated.

SPARE PARTS - Use only new genuine Miller parts for replacement or repair.

PRODUCT LABELS – Per ANSI 535.4-1991 labels should be inspected and cleaned and safety related labels replaced when no longer legible. Contact Miller for replacements.

DIMENSIONS in this catalog are generally nominal dimensions in inches unless otherwise indicated. Where tolerance information is needed but is not indicated please contact Miller for applicable tolerances.

WEIGHTS in this catalog are generally in pounds unless otherwise indicated.

MILLER SWIVELS- TYPES, USE, SELECTION

With the largest product line of load handling swivels worldwide, and continuous leadership and innovation in the use of swivels since our founding in 1935 in Pomona, California, Miller provides users worldwide with proven solutions to everyday problems in the field, under the sea, in the air, in the factory and in space. Miller also has provided many custom swivel solutions tailored to specific operations in the construction, mining, petroleum and natural gas, foundation drilling, nuclear, communications and other industries.

The five general categories of Miller Swivels are:

- Y-Link, angular contact ball bearing swivels
- Econo-Link, tapered thrust bearing swivels
- Special application swivels (marine, stainless, insulated, demolition, drilling, line pulling)
- Custom-designed swivels for specific operational requirements

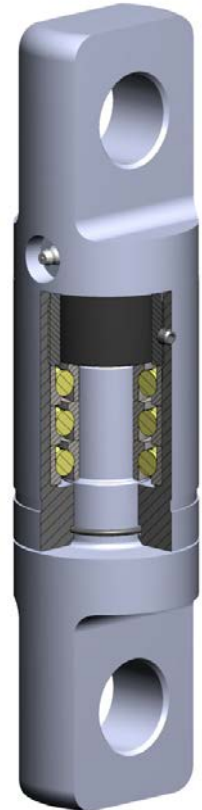
Y-Link Angular Contact Ball Bearing Swivels

Developed by Miller, angular contact ball bearing swivels provide the most efficient way to handle the heavy thrust loads encountered in crane operation and other lifting and pulling jobs. The precision-ground, precisely matched bearings assure longer service life, superior performance, and much longer wire rope life where rope twist must be reduced or eliminated. The thrust load is transferred from an overlapping shoulder on the bearing race through the balls to the diagonally opposed shoulder on the outer bearing race. This distributes the load to the outer part of the barrel.

The angular contact bearings are mounted in complements of two or more, depending on the load capacity of the swivel. They permit the load to rotate freely, yet they prevent the load from spinning. By permitting the wire rope to neutralize itself, they can extend the wire rope life multiple times.

The angular contact ball bearing swivel design results in highly reduced torque levels- 50% to 70% less than the torque levels found for traditional thrust bearing swivels, especially as load increases.

External parts of Miller Y-Link Angular Contact Ball bearings are machined from high quality alloy steel and are zinc plated for long-term corrosion protection. Working loads range from ½ to 1000 tons.



Econo-Link Thrust Bearing Swivels

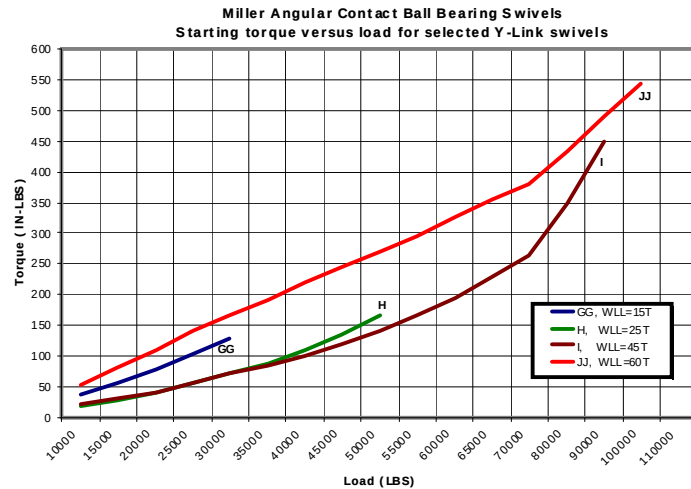
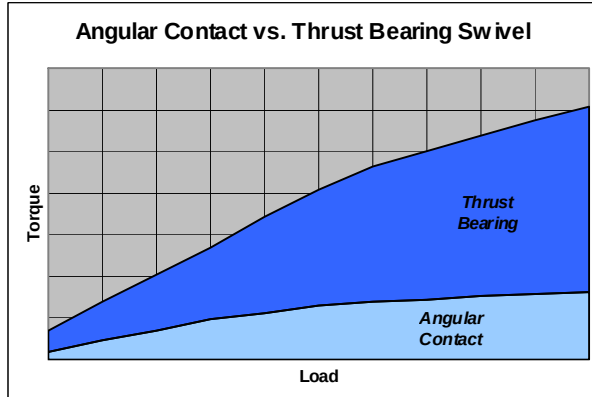


Outstanding economy and good service are the hallmarks of Miller's traditional swivel model. Miller thrust bearing swivels are economically priced, yet are surpassed only by Miller's angular contact bearing swivels. Miller Econo-Link Thrust Bearing Swivels incorporate a single heavy duty tapered roller bearing housed in a steel retainer. External parts are steel castings with forged steel hooks when hooks are used. A close-fitting bronze bushing seals lubricant in and keeps dirt out of the bearing. For many applications Miller Econo-Link Thrust Bearing Swivels offer the most economical, yet dependable, service available. Working loads range from 3 to 15 tons.

MILLER SWIVELS- TYPES, USE, SELECTION

Torque Considerations

Angular contact bearings allow much lower torque resistance. Starting torque under load for angular contact bearings is based on Miller testing and empirical data.



"General guide only, torque value may vary to a significant degree due to application conditions. Miller does not guarantee torque data shown, should specific torque performance be desired, contact Miller"

SPECIAL APPLICATION SWIVELS with Angular Contact Ball Bearings

Over the years Miller has developed a number of angular contact ball bearing swivels for given applications or environments. The following are the most common and those for which we have developed standard products available to order from Miller. All of these swivels are presented in the tables following in this section of the catalog.

Heavy Duty Swivels (HD-Link) – These swivels which feature heavy duty barrels are available to protect bearings and maximize swivel life under the heavy impact and abrasive environment of demolition work. Many other Miller swivels can be fitted with heavy duty barrels upon request.

Overhaul Swivels (g-Link) – Weighing from 2.5 to nearly 4 times as much as standard swivels of the same type, these special Miller Overhaul Swivels provide an excellent means of adding additional weight without increasing overall dimensions. Ideal for small truck cranes and hoists in applications where minimal additional overhaul weight is required.

Line Pulling Swivels (Miller-Pengo) – Designed specifically by Miller for the electrical distribution industry for stringing high tension wires and for pulling wires through conduit. They help to accommodate the inevitable twisting developed as lines are installed using sheaves over long distances. These swivels are not designed for general lifting applications. Please see Miller Y-Link and Econo-Link for lifting swivels.

Stainless Steel Swivels – These swivels are designed for cable and chain applications above the surface of the water where long term environmental considerations require the corrosion resistant qualities of stainless steel. In addition to above surface marine, other typical applications include chemical processing plants, nuclear plants, and applications with magnetic permeability restrictions.

MILLER SWIVELS- TYPES, USE, SELECTION

Stainless Steel, Underwater High Pressure Swivels – A unique pressure compensating, high pressure seal allows this series of swivels to operate reliably up to depths of 10,000 feet. These swivels find applications in oceanographic work, offshore mooring, cable laying, offshore oil and gas exploration, installation, operating activities, commercial fishing, defense and other marine applications. High pressure swivels are also available in non-stainless steel.

Insulated Swivels – Miller also produces a series of insulated swivels for working loads up to 120 tons. These swivel assemblies incorporate the Miller ISO/Link-AC or ISO/Link-DC for the personnel safety purposes of isolating the load from high voltage electrical current in case of boom contact with power lines or to prevent stray building grid currents in industrial applications where welding or induction melting occurs. Please refer to the Load Insulating section of the catalog for information on these high value swivels.

MILLER SWIVEL SELECTION GUIDE

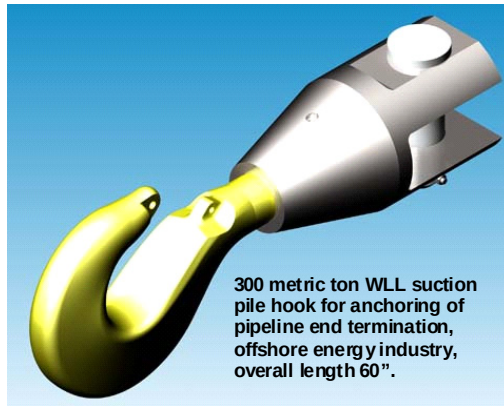
SWIVEL MODEL	Y-Link	Y-Link/HP	Hydro-Link	Hydro-Link /HP	Econo-Link
DESIGN FACTOR	5 for WLL ≤ 35T • 4 for WLL > 35T				4
WLL RANGE*	.45 to 300T	.45 to 300T	.45 to 300T	.45 to 300T	3 to 15T
MBL RANGE	N/A	N/A	N/A	N/A	N/A
BEARING TYPE	Angular Contact	Angular Contact	Angular Contact	Angular Contact	Tapered Thrust
SEAL TYPE	O-ring	Pressure Compensating Spring-Energized Polymer	O-ring	Pressure Compensating Spring-Energized Polymer	Bronze Bushing
FEATURES	Machined Alloy Steel, Best All Around, 54 Configurations	Y-Link, Sealed for High Pressure Applications	Machined Stainless Steel Construction	Hydro-Link, Sealed for High Pressure	Cast Alloy Steel, Low Price, 6 Popular Configurations
LUBRICATION	Grease filled, E-Size and Up Include Zerk	Hot Poured Miller Lube, Oil Plug	Grease filled, E-Size and Up Include Zerk	Hot Poured Miller Lube, Oil Plug	Grease filled, No Zerk
REBUILDABLE	Yes	Yes	Yes	Yes	No
APPLICATIONS	Full Range Lifting, Custom End Configurations, Low Friction	Short Term Sub Sea to Depths of 10,000 ft, Offshore Oil and Gas	Above Surface Marine, Chemical and Nuclear Plants	Long Term Sub Sea to Depths of 10,000 ft, Offshore Oil and Gas	General lifting, Economical
SWIVEL MODEL	Econo Swiveler	g-Link	HD-Link	Pengo / Miller	
DESIGN FACTOR	4	5	5	3	
WLL RANGE*	3 to 15T	1.5 to 8.5T	5 to 35T	.9 to 50T	
MBL RANGE	N/A	N/A	N/A	N/A	
BEARING TYPE	Tapered Thrust	Angular Contact	Angular Contact	Angular Contact	
SEAL TYPE	Bronze Bushing	O-ring	O-ring	O-ring	
FEATURES	2 Configurations, 35 - 575 lbs Overhaul Weight	2.5 to 4 Times Overhaul weight Compared to Y-Links	Heavy Duty Barrel for Increased Bearing Protection and Service Life	For Tension Line Pulling, 90 Degree Connectors Available.	
LUBRICATION	Grease filled, No Zerk	Grease Filled, D-Size and Up have Zerk	Grease Filled, E-Size and Up have Zerk	Grease filled, No Zerk	
REBUILDABLE	No	Yes	Yes	No	
APPLICATIONS	Small Cranes, Low Head Room, Overhaul Weight	Small Truck Cranes, Low Head Room, Minimal Overhaul Weight	Demolition and High Impact	Electrical Distribution, Line Stringing / Pulling	
* For standard swivels. Custom swivel working loads available to 1000 tons					

MILLER CUSTOM SWIVELS

Miller has designed and manufactured swivels for specific high value lifting or handling projects in various industries including construction, offshore energy, aerospace and defense, nuclear and others. Following are just a few examples. Please contact us directly for information on custom swivel products.



375 ton WLL narrow body with side load capability, for riser pull-in, offshore energy industry. 65"L x Ø13"



300 metric ton WLL suction pile hook for anchoring of pipeline end termination, offshore energy industry, overall length 60".



125T swivel, military transport



Insulated, 60T, NASA shuttle maintenance operations



375T offshore special for SCR pull-in

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS



- First ever angular contact bearing swivel, USA patent 2,651,533
- Working Load Limits from 0.45 through 300 tons
- Multiple Angular Contact Bearing Design extends the life of the wire rope, and swivel life
- Low starting torque enables immediate swivel response
- Angular contact bearings endure higher rotational speeds
- Models 5 tons and up include lubrication fitting
- Durable structural parts are made from forged alloy steel- not cast
- Zinc plated inside and out for effective protection against corrosion
- Design Factor is 5:1 through 35 tons and 4:1 above 35 tons
- Proof Load is 2 x WLL
- See table of contents for stainless steel, high pressure, and other special models

Miller Swivel Products

Largest, most comprehensive inventory for immediate shipment in North America

- Y-Link, the original angular contact ball bearing swivel
- Rugged, economical Econo-Link thrust bearing swivels
- Overhaul ball assemblies for every lift height
- Custom swivel designs for specific operational requirements (e.g., drilling, stainless, offshore, line pull, mooring, insulated)

Y-Link, Angular Contact Ball Bearing



Econo-Link, Tapered Roller Thrust Bearing



Overhaul Ball Assemblies



Special Requirements



High Quality Lifting Products since 1935

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SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

Y-LINK MODEL NUMBER SYSTEM

M O D E L	W L L	CAPS									BARRELS	SHANKS					EYE BOLT	
		CLEVIS	EYE	STUD	BULLET NOSE	SOCKET	THIMBLE	WEDGE	HOOK	EYE BOLT	STANDARD BARREL	CLEVIS	EYE	STUD	BULLET NOSE	HOOK		
A	0.5	✓	✓	✓	✓	5	6	7	8	9	1	1	2	3	4	5	6	
BB	0.8	✓	✓	✓	✓						✓	✓	✓	✓	✓			
B	1.5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
C	3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
D	5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
EE	8.5	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
G	10	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
GG	15	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
H	25	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
HH	35	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
I	45	✓	✓	✓	✓						✓	✓			✓			
J	50	✓	✓	✓							✓	✓	✓		✓	✓		
JJ	60	✓	✓	✓							✓	✓	✓		✓	✓		
K	75	✓	✓	✓								✓	✓		✓	✓		
KK	100	✓	✓	✓							✓	✓	✓		✓	✓		
KKK	150	✓	✓								✓	✓	✓		✓	✓		
L	200	✓	✓									✓	✓			✓		
M	300	✓	✓								✓	✓	✓			✓		
MODEL		1	2	3	4	5	6	7	8	9	1	1	2	3	4	5	6	
CAPS												SHANKS						
												BARRELS						

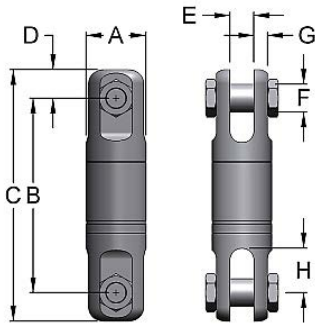
NOTE: ☒ CHECKED COMPONENTS ARE AVAILABLE AS STANDARD PRODUCTS. OTHERS MAY BE ADDED. FOR UNCHECKED, PLEASE CONTACT MILLER

MODEL NUMBER KEY:			
X	#	#	#
MODEL LETTER(S)	CAP NUMBER	BARREL NUMBER	SHANK NUMBER

EXAMPLE: D-115 = (D) 5 TON, (1) CLEVIS CAP, (1) STD. BARREL, (5) HOOK

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

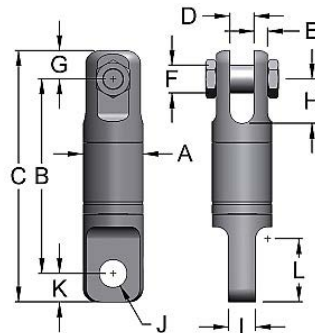
CLEVIS TO CLEVIS - TYPE 111



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
0.45	A-111	1/8	0.88	2.38	3.13	0.38	0.25	0.25	0.19	0.41	0.37
0.75	BB-111	1/4	1.31	3.5	4.38	0.44	0.31	0.38	0.22	0.56	0.93
1.5	B-111	3/8	1.63	4.06	5.44	0.69	0.5	0.5	0.31	0.78	2
3	C-111	1/2	2	6.25	8.09	0.94	0.75	0.75	0.38	1.19	4.5
5	D-111	5/8	2.5	7.94	10.09	1.13	1	0.88	0.56	1.56	9.5
8.5	EE-111	3/4	2.88	9.63	12.31	1.34	1.56	1.19	0.53	2.09	15.25
10	G-111	7/8	4	14	17.5	1.75	1.75	1.5	0.81	3.5	40
15	GG-111	1	4	14	17.5	1.75	1.75	1.5	0.81	3.5	40
25	H-111	1 1/4	5	15.94	20.69	2.38	2	2	1.13	3.69	78
35	HH-111	1 1/2	5	15.94	20.69	2.38	2	2	1.13	3.69	78
45	I-111		6	20.06	26.06	3	2.5	2.25	1.25	4	146
50	J-111		7	20.94	27.94	3.5	3	2.5	1.5	4.5	207
60	JJ-111		7	20.94	27.94	3.5	3	2.5	1.5	4.5	207
75	K-111		8	24	32	4	3.5	3	1.75	5	300
100	KK-111		10	32.88	43.88	5.5	4.06	4	2	6.75	692
150	KKK-111		10	32.88	43.88	5.5	4.06	4	2	6.75	692
200	L-111		11.5	36.52	48.02	5.75	5.06	4.5	2.56	7.75	985
300	M-111		14	43.95	58.95	7.5	6.13	4.75	3.94	9.5	1680

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

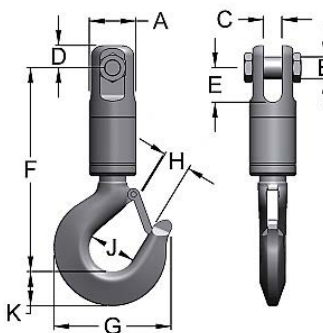
CLEVIS TO EYE - TYPE 112



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.45	A-112	1/8	0.88	2.5	3.25	0.25	0.19	0.25	0.38	0.41	0.25	.26	0.38	0.75	0.31
0.75	BB-112	1/4	1.31	3.56	4.44	0.31	0.22	0.38	0.44	0.56	0.31	0.38	0.44	0.94	0.94
1.5	B-112	3/8	1.63	4.13	5.38	0.5	0.31	0.5	0.69	0.78	0.5	0.66	0.63	1.34	1.94
3	C-112	1/2	2	6.19	8.13	0.75	0.38	0.75	0.94	1.19	0.75	0.91	1	1.94	4.25
5	D-112	5/8	2.5	7.88	10.19	1	0.56	0.88	1.13	1.56	1	1.28	1.19	2.5	8.94
8.5	EE-112	3/4	2.88	9.5	12.34	1.56	0.53	1.19	1.34	2.19	1.25	1.41	1.5	3.13	15.25
10	G-112	7/8	4	13.94	17.5	1.75	0.81	1.5	1.75	3.5	1.72	1.66	1.81	4.66	39
15	GG-112	1	4	13.63	17.5	1.75	0.81	1.5	1.75	3.5	1.94	2.03	2.13	4.91	40
25	H-112	1 1/4	5	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
35	HH-112	1 1/2	5	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
45	I-112		6	20.06	26.06	2.5	1.25	2.25	3	4	2.5	2.53	3	7	139
50	J-112		7	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	205
60	JJ-112		7	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	205
75	K-112		8	24	32	3.5	1.75	3	4	5	3.5	3.38	4	8.13	305
100	KK-112		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	647
150	KKK-112		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.28	5.5	11.38	647
200	L-112		11.5	35.52	47.27	5.06	2.56	4.5	5.75	7.75	5	4.81	6	13	925
300	M-112		14	43.45	58.95	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

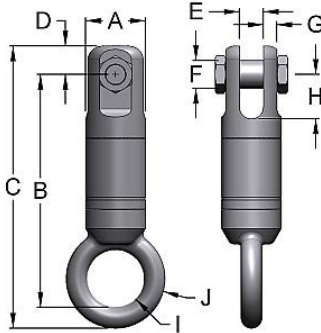
CLEVIS TO HOOK - TYPE 115



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	J	K	Weight Lbs.
1.5	B-115	3/8	1.63	0.5	0.5	0.69	0.78	5.88	3.63	1	1.63	1	2.31
3	C-115	1/2	2	0.75	0.75	0.94	1.19	8.56	4.88	1.34	2	1.38	5.94
5	D-115	5/8	2.5	0.88	1	1.13	1.56	10.81	6.5	1.34	2.5	1.81	11.81
8.5	EE-115	3/4	2.88	1.19	1.56	1.34	2.09	13.75	8.69	2.25	3.25	2.59	24
10	G-115	7/8	4	1.5	1.75	1.75	3.5	17.88	11	3	3.25	3	53
15	GG-115	1	4	1.5	1.75	1.75	3.5	17.88	11	3	4.25	3	53
25	H-115	1 1/4	5	2	2	2.38	3.69	23.01	12.02	3.54	4.41	4.21	131
35	HH-115	1 1/2	5	2	2	2.38	3.69	23.01	12.02	3.54	4.41	4.21	131
45	I-115		6	2.25	2.5	3	4	27.65	13.46	3.94	4.92	4.69	211
50	J-115		7	2.5	3	3.5	4.5	29.34	15.12	4.41	5.51	5.26	293
60	JJ-115		7	2.5	3	3.5	4.5	29.34	15.12	4.41	5.51	5.26	293
75	K-115		8	3	3.5	4	5	33.36	17	4.92	6.3	5.97	411
100	KK-115		10	4	4.06	5.5	6.75	45.31	21.34	6.3	7.87	7.55	922
150	KKK-115		10	4	4.06	5.5	6.75	45.31	21.34	6.3	7.87	7.5	907
200	L-115		11.5	4.5	5.06	5.75	7.75	50.48	26.95	7.87	9.84	9.29	1510
300	M-115		14	4.75	6.13	7.5	9.5	63.13	33.69	9.17	12.4	11.81	2800

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

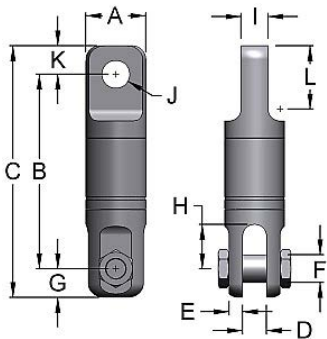
CLEVIS TO EYE BOLT - TYPE 116



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	Weight Lbs.
1.5	B-116	3/8	1.63	5.44	6.75	0.69	0.5	0.5	0.31	0.78	1.5	2.75	2.19
3	C-116	1/2	2	7.47	9.16	0.94	0.75	0.75	0.38	1.19	1.75	3.5	4.5
5	D-116	5/8	2.5	9.31	11.44	1.13	1	0.88	0.56	1.56	2	4	9.56
8.5	EE-116	3/4	2.88	11.09	13.44	1.34	1.56	1.19	0.53	2.09	2.5	4.5	14.75
10	G-116	7/8	4	15.91	19.47	1.75	1.75	1.5	0.81	3.5	3.25	6.88	43
15	GG-116	1	4	15.91	19.47	1.75	1.75	1.5	0.81	3.5	3.25	6.88	43

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

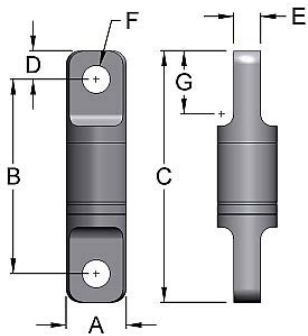
EYE TO CLEVIS - TYPE 211



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.45	A-211	1/8	0.88	2.5	3.25	0.25	0.19	0.25	0.38	0.41	0.25	.26	0.38	0.75	0.38
0.75	BB-211	1/4	1.31	3.63	4.5	0.31	0.22	0.38	0.44	0.44	0.31	.39	0.44	0.94	0.94
1.5	B-211	3/8	1.63	4.13	5.44	0.5	0.31	0.5	0.69	0.78	0.5	0.66	0.63	1.34	1.88
3	C-211	1/2	2	6.19	8.13	0.75	0.38	0.75	0.94	1.19	0.75	0.91	1	1.94	4.31
5	D-211	5/8	2.5	7.88	10.19	1	0.56	0.88	1.13	1.56	1	1.28	1.19	2.5	9.06
8.5	EE-211	3/4	2.88	9.5	12.34	1.56	0.53	1.19	1.34	2.19	1.25	1.41	1.5	3.13	14.75
10	G-211	7/8	4	13.94	17.5	1.75	0.81	1.5	1.75	3.5	1.72	1.66	1.81	4.66	39
15	GG-211	1	4	13.63	17.5	1.75	0.81	1.5	1.75	3.5	1.94	2.03	2.13	4.91	40
25	H-211	1 1/4	5	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
35	HH-211	1 1/2	5	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
45	I-211		6	20.06	26.06	2.5	1.25	2.25	3	4	2.5	2.53	3	7	139
50	J-211		7	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	211
60	JJ-211		7	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	211
75	K-211		8	24	32	3.5	1.75	3	4	5	3.5	3.38	4	8.13	305
100	KK-211		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	678
150	KKK-211		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.28	5.5	11.38	678
200	L-211		11.5	36.67	48.42	5.06	2.56	4.5	5.75	7.75	5	4.81	6	13	960
300	M-211		14	43.45	58.95	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

EYE TO EYE - TYPE 212

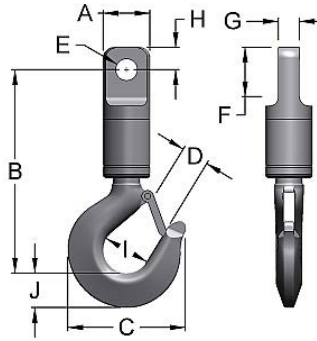


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.45	A-212	1/8	0.88	2.63	3.38	0.38	0.25	.26	0.75	0.31
0.75	BB-212	1/4	1.31	3.81	4.69	0.44	0.31	.39	0.94	0.93
1.5	B-212	3/8	1.63	4.13	5.38	0.63	0.5	0.66	1.34	1.87
3	C-212	1/2	2	6.13	8.13	1	0.75	0.91	1.94	4.06
5	D-212	5/8	2.5	7.81	10.19	1.19	1	1.28	2.5	8.5
8.5	EE-212	3/4	2.88	9.31	12.31	1.5	1.25	1.41	3.13	14.75
10	G-212	7/8	4	13.88	17.5	1.81	1.72	1.66	4.66	37
15	GG-212	1	4	13.25	17.5	2.13	1.94	2.03	4.91	39
25	H-212	1 1/4	5	15.94	20.69	2.38	2.25	2.31	5.25	72
35	HH-212	1 1/2	5	15.94	20.69	2.38	2.25	2.31	5.25	72
45	I-212		6	20.06	26.06	3	2.5	2.53	7	132
50	J-212		7	20.94	27.94	3.5	2.94	2.88	7	209
60	JJ-212		7	20.94	27.94	3.5	2.94	2.88	7	209
75	K-212		8	24	32	4	3.5	3.38	8.13	311
100	KK-212		10	32.88	43.88	5.5	4.5	4.03	11.38	653
150	KKK-212		10	32.88	43.88	5.5	4.5	4.28	11.38	657
200	L-212		11.5	35.67	47.67	6	5	4.81	13	900
300	M-212		14	42.95	58.95	8	6	6.06	17	1610

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

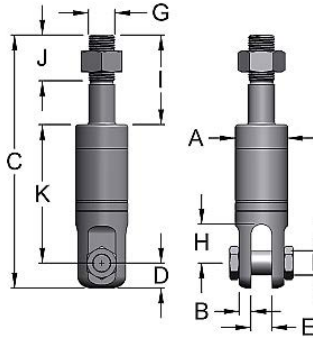
EYE TO HOOK - TYPE 215



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	Weight Lbs.
1.5	B-215	3/8	1.63	6.19	3.63	1	0.66	1.34	0.5	0.63	1.63	1	2.25
3	C-215	1/2	2	8.56	4.88	1.34	0.91	1.94	0.75	1	2	1.38	5.81
5	D-215	5/8	2.5	10.88	6.5	1.69	1.28	2.5	1	1.19	2.5	1.81	11.37
8.5	EE-215	3/4	2.88	13.63	8.69	2.25	1.41	3.13	1.25	1.5	3.25	2.59	23.5
10	G-215	7/8	4	17.81	11	3	1.66	4.66	1.72	1.81	3.25	3	52
15	GG-215	1	4	17.56	11	3	2.03	4.91	1.94	2.13	4.25	3	53
25	H-215	1 1/4	5	23.01	12.02	3.54	2.31	5.25	2.25	2.38	4.41	4.21	128
35	HH-215	1 1/2	5	23.01	12.02	3.54	2.31	5.25	2.25	2.38	4.41	4.21	128
45	I-215		6	27.65	13.46	3.94	2.53	7	2.5	3	4.92	4.69	205
50	J-215		7	29.34	15.12	4.41	2.88	7	2.94	3.5	5.51	5.26	289
60	JJ-215		7	29.34	15.12	4.41	2.88	7	2.94	3.5	5.51	5.26	289
75	K-215		8	33.36	17	4.92	3.38	8.13	3.5	4	6.3	5.97	423
100	KK-215		10	45.31	21.34	6.3	4.03	11.38	4.5	5.5	7.87	7.55	913
150	KKK-215		10	45.31	21.34	6.3	4.28	11.38	4.5	5.5	7.87	7.55	911
200	L-215		11.5	50.82	26.95	7.87	4.81	13	5	6	9.84	9.29	1485
300	M-215		14	63.13	33.69	9.17	6.06	17	6	8	12.4	8.63	2835

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

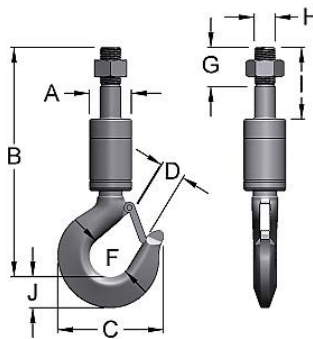
STUD TO CLEVIS - TYPE 311



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	Weight Lbs.
0.45	A-311	1/8	0.88	0.19	4.06	0.38	0.25	0.25	0.38	0.41	1.5	1.19	2.19	0.37
0.75	BB-311	1/4	1.31	0.22	5.63	0.44	0.31	0.38	0.63	0.56	2	1.5	3.19	1.06
1.5	B-311	3/8	1.63	0.31	6.06	0.69	0.5	0.5	0.75	0.78	2	1.5	3.75	1.87
3	C-311	1/2	2	0.38	9.19	0.94	0.75	0.75	0.88	1.19	3	2.63	5.25	4.5
5	D-311	5/8	2.5	0.56	11.19	1.13	1	0.88	1.25	1.56	3.5	3.13	6.56	9.75
8.5	EE-311	3/4	2.88	0.53	13.03	1.56	1.56	1.19	1.5	2.19	4	3	7.69	14.5
10	G-311	7/8	4	0.81	17.44	1.75	1.75	1.5	2	3.5	5.5	3	10.19	37
15	GG-311	1	4	0.81	17.44	1.75	1.75	1.5	2	3.5	5.5	3	10.19	37
25	H-311	1 1/4	5	1.13	23.31	2.38	2	2	2.5	3.69	8.5	4.25	12.44	78
35	HH-311	1 1/2	5	1.13	23.31	2.38	2	2	2.5	3.69	8.5	4.25	10.44	78
45	I-311		6	1.25	25.56	3	2.5	2.25	3	4	8	4	14.56	131
50	J-311		7	1.5	28.44	3.5	3	2.5	3.25	4.5	8	4	16.94	198
60	JJ-311		7	1.5	28.44	3.5	3	2.5	3.25	4.5	8	4	16.94	198
75	K-311		8	1.75	33.5	4	3.5	3	3.75	5	10	5	19.5	271
100	KK-311		10	2	51.13	5.5	4.06	4	6.5	6.75	20.5	7.25	25.13	648

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

STUD TO HOOK - TYPE 315

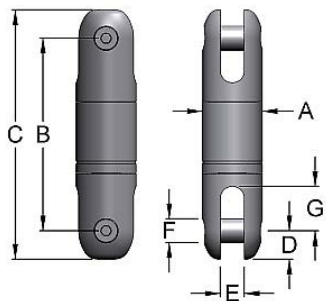


WLL* Tons	Model Number	Rope Size	A	B	C	D	F	G	H	I	J	Weight Lbs.
1.5	B-315	3/8	1.63	7.25	3.63	1	1.63	1.5	0.75	2	1	2.25
3	C-315	1/2	2	10.63	4.88	1.34	2	2.63	0.88	3	1.38	6
5	D-315	5/8	2.5	13	6.5	1.69	2.5	3.13	1.25	3.5	1.81	12.06
8.5	EE-315	3/4	2.88	15.81	8.69	2.25	3.25	3	1.5	4	2.59	23.25
10	G-315	7/8	4	19.59	11	3	3.25	3	2	5.5	3	50
15	GG-315	1	4	19.59	11	3	4.25	3	2	5.5	3	50
25	H-315	1 1/4	5	28.5	12.02	3.54	4.41	4.25	2.5	8.5	4.21	133
35	HH-315	1 1/2	5	28.5	12.02	3.54	4.41	4.25	2.5	8.5	4.21	133
45	I-315		6	30.13	13.46	3.94	4.92	4	3	8	4.69	188
50	J-315		7	33.34	15.12	4.41	5.51	4	3.25	8	5.26	278
60	JJ-315		7	33.34	15.12	4.41	5.51	4	3.25	8	5.26	278
75	K-315		8	38.82	17	4.92	6.3	5	3.75	10	5.97	403
100	KK-315		10	58.2	21.34	6.3	7.87	7.25	6.5	20.5	7.55	941

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

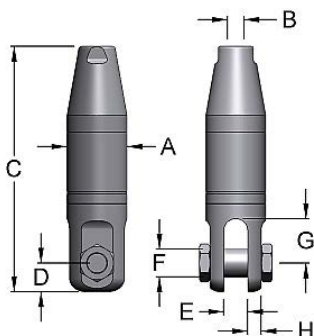
BULLET NOSE TO BULLET NOSE - TYPE 414



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.45	A-414	1/8	0.88	2.38	3.13	0.38	0.25	0.31	0.41	0.37
0.75	BB-414	1/4	1.31	3.56	4.44	0.44	0.31	0.38	0.56	1.12
1.5	B-414	3/8	1.63	4.06	5.19	0.56	0.5	0.44	0.81	1.81
3	C-414	1/2	2	5.44	7.06	0.81	0.75	0.63	0.94	3.87
5	D-414	5/8	2.5	7.88	10.19	1.13	1	0.88	1.56	8.06
8.5	EE-414	3/4	2.88	9.81	12.31	1.25	1.28	1	2.13	14.5
10	G-414	7/8	4	13.25	16.75	1.75	1.75	1.5	3.25	40
15	GG-414	1	4	13.25	16.75	1.75	1.75	1.5	3.25	40
25	H-414	1 1/4	5	15.91	20.66	2.38	2	2	3.69	84
35	HH-414	1 1/2	5	15.91	20.66	2.38	2	2	3.69	84
45	I-414		6	20.08	26.08	3	2.5	2.25	4	134

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

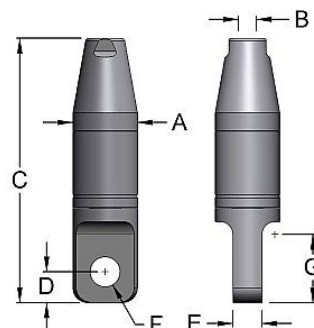
SOCKET TO CLEVIS - TYPE 511



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
1.5	B-511	3/8	1.63	0.44	6.19	0.69	0.5	0.5	0.78	0.31	1.87
3	C-511	1/2	2	0.56	8.94	0.94	0.75	0.75	1.19	0.38	4.43
5	D-511	5/8	2.5	0.69	10.44	1.13	1	0.88	1.56	0.56	8.68
8.5	EE-511	3/4	2.88	0.81	12.75	1.34	1.56	1.19	2.09	0.53	14.5
10	G-511	7/8	4	0.97	16.25	1.75	1.75	1.5	3.5	0.81	33
15	GG-511	1	4	1.13	16.25	1.75	1.75	1.5	3.5	0.81	33
25	H-511	1 1/4	5	1.38	20.19	2.38	2	2	3.69	1.13	74
35	HH-511	1 1/2	5	1.63	20.19	2.38	2	2	3.69	1.13	74

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

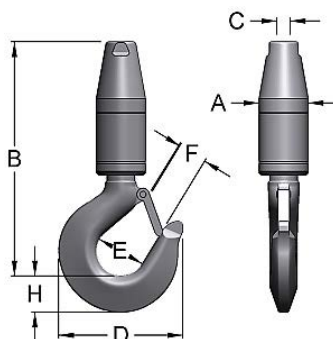
SOCKET TO EYE - TYPE 512



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
1.5	B-512	3/8	1.63	0.44	6.19	0.63	0.5	0.66	1.87
3	C-512	1/2	2	0.56	8.94	1	0.75	0.91	4.43
5	D-512	5/8	2.5	0.69	10.44	1.19	1	1.28	8.68
8.5	EE-512	3/4	2.88	0.81	12.75	1.5	1.25	1.41	14.5
10	G-512	7/8	4	0.97	16.25	1.81	1.72	1.66	33
15	GG-512	1	4	1.13	16.25	2.13	1.94	2.03	33
25	H-512	1 1/4	5	1.38	20.19	2.38	2.25	2.31	74
35	HH-512	1 1/2	5	1.63	20.19	2.38	2.25	2.31	74

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SOCKET TO HOOK - TYPE 515

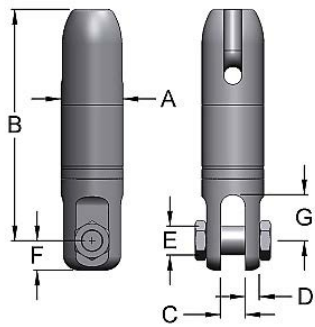


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	H	Weight Lbs.
1.5	B-515	3/8	1.63	7.38	0.44	3.63	1.63	1	1	2.18
3	C-515	1/2	2	10.38	0.56	4.88	2	1.34	1.38	5.87
5	D-515	5/8	2.5	12.19	0.69	6.5	2.5	1.69	1.81	11
8.5	EE-515	3/4	2.88	15.56	0.81	8.69	3.25	2.25	2.59	23.25
10	G-515	7/8	4	18.38	0.97	11	3.25	3	3	46
15	GG-515	1	4	18.38	1.13	11	4.25	3	3	46
25	H-515	1 1/4	5	24.88	1.38	12.02	4.41	3.54	4.21	125
35	HH-515	1 1/2	5	24.88	1.63	12.02	4.41	3.54	4.21	125

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

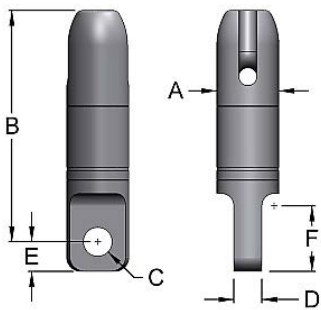
THIMBLE TO CLEVIS - TYPE 611



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
1.5	B-611	3/8	1.63	6	0.5	0.31	0.5	0.69	0.78	2.5
3	C-611	1/2	2	8.06	0.75	0.38	0.75	0.94	1.19	5.25
5	D-611	5/8	2.5	9.31	1	0.56	0.88	1.13	1.56	10
8.5	EE-611	3/4	2.88	11.38	1.56	0.53	1.19	1.34	2.09	16.5
10	G-611	7/8	4	15.38	1.75	0.81	1.5	1.75	3.5	42
15	GG-611	1	4	15.38	1.75	0.81	1.5	1.75	3.5	42
25	H-611	1 1/4	5	18.31	2	1.13	2	2.38	2.69	80
35	HH-611	1 1/2	5	18.31	2	1.13	2	2.38	2.69	80

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

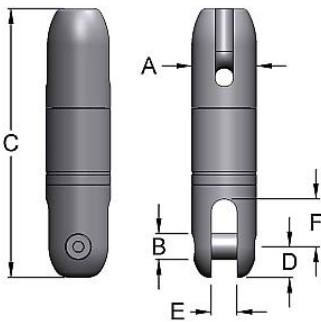
THIMBLE TO EYE - TYPE 612



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
1.5	B-612	3/8	1.63	6.13	0.66	0.5	0.63	1.34	2.43
3	C-612	1/2	2	8	0.91	0.75	1	1.94	6
5	D-612	5/8	2.5	9.25	1.28	1	1.19	2.5	9.43
8.5	EE-612	3/4	2.88	11.31	1.41	1.25	1.5	3.13	16.5
10	G-612	7/8	4	15.44	1.66	1.72	1.81	4.66	41
15	GG-612	1	4	15.44	2.03	1.94	2.13	4.91	42
25	H-612	1 1/4	5	18.31	2.31	2.25	2.38	5.25	75
35	HH-612	1 1/2	5	18.31	2.31	2.25	2.38	5.25	75

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

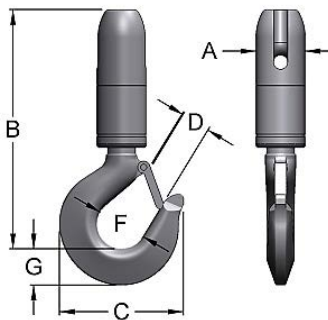
THIMBLE TO BULLET NOSE - TYPE 614



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
1.5	B-614	3/8	1.63	0.44	6.61	1	0.5	1.5	2
3	C-614	1/2	2	0.63	8.55	1.34	0.75	2.63	6.06
5	D-614	5/8	2.5	0.88	10.44	1.69	1	3.13	9.5
8.5	EE-614	3/4	2.88	1	12.77	1.25	1.28	2.13	16.5
10	G-614	7/8	4	1.5	17.25	3	1.75	3	38.5
15	GG-614	1	4	1.5	17.25	3	1.75	3	38.5
25	H-614	1 1/4	5	2	20.69	3.63	2	4.25	76
35	HH-614	1 1/2	5	2	20.69	3.75	2	4.25	76

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

THIMBLE TO HOOK - TYPE 615

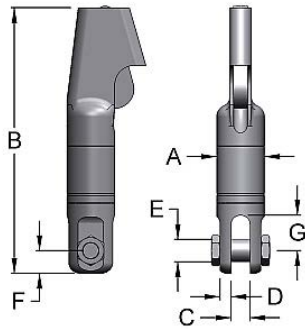


WLL* Tons	Model Number	Rope Size	A	B	C	D	F	G	Weight Lbs.
1.5	B-615	3/8	1.63	7.81	3.63	1	1.63	1	2.87
3	C-615	1/2	2	10.44	4.88	1.34	2	1.38	6.81
5	D-615	5/8	2.5	12.19	6.5	1.69	2.5	1.81	12.31
8.5	EE-615	3/4	2.88	15.56	8.69	2.25	3	2.59	25.25
10	G-615	7/8	4	19.38	11	3	3.25	3	55
15	GG-615	1	4	19.38	11	3	4.25	3	55
25	H-615	1 1/4	5	25.38	12.02	3.54	4.41	4.21	136
35	HH-615	1 1/2	5	25.38	12.02	3.54	4.41	4.21	136

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK, ANGULAR CONTACT BEARINGS

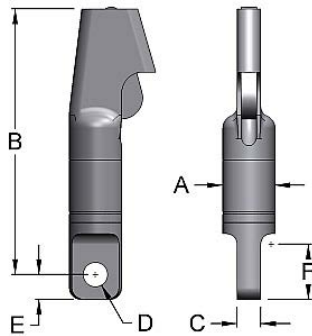
WEDGE TO CLEVIS - TYPE 711



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
3	C-711	1/2	2	10.25	0.75	0.38	0.75	0.94	1.19	6.31
5	D-711	1/2, 5/8	2.5	12.53	1	0.56	0.88	1.13	1.56	12.56
8.5	EE-711	1/2, 5/8, 3/4	2.88	14.44	1.56	0.53	1.19	1.34	2.09	20.5
10	G-711	3/4, 7/8, 1	4	19.75	1.75	0.81	1.5	1.75	3.5	47
15	GG-711	3/4, 7/8, 1	4	19.75	1.75	0.81	1.5	1.75	3.5	47
25	H-711	1 1/8, 1 1/4	5	27.31	2	1.13	2	2.38	3.69	98
35	HH-711	1 1/8, 1 1/4	5	27.31	2	1.13	2	2.38	3.69	98

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

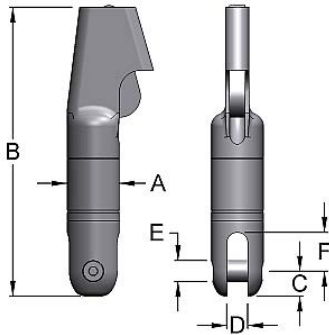
WEDGE TO EYE - TYPE 712



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
3	C-712	1/2	2	10.25	0.75	0.91	1	1.94	6.06
5	D-712	1/2, 5/8	2.5	12.5	1	1.28	1.19	2.5	12
8.5	EE-712	1/2, 5/8, 3/4	2.88	14.25	1.25	1.41	1.5	3.13	20.5
10	G-712	3/4, 7/8, 1	4	19.69	1.72	1.66	1.81	4.66	45
15	GG-712	3/4, 7/8, 1	4	19.38	1.94	2.03	2.13	4.91	46
25	H-712	1 1/8, 1 1/4	5	25.43	2.25	2.31	2.38	5.25	94
35	HH-712	1 1/8, 1 1/4	5	25.43	2.25	2.31	2.38	5.25	94

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

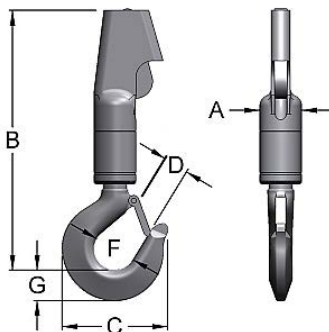
WEDGE TO BULLET NOSE - TYPE 714



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
3	C-714	1/2	2	9.92	0.81	0.75	0.63	0.94	3.87
5	D-714	1/2, 5/8	2.5	12.42	1.13	1	0.88	1.56	8.06
8.5	EE-714	1/2, 5/8, 3/4	2.88	14.47	1.25	1.28	1	2.13	14.5
10	G-714	3/4, 7/8, 1	4	19.75	1.75	1.75	1.5	3.25	40
15	GG-714	3/4, 7/8, 1	4	19.75	1.75	1.75	1.5	3.25	40
25	H-714	1 1/8, 1 1/4	5	27.3	2.38	2	2	3.69	96
35	HH-714	1 1/8, 1 1/4	5	27.3	2.38	2	2	3.69	96

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

WEDGE TO HOOK - TYPE 715



WLL* Tons	Model Number	Rope Size	A	B	C	D	F	G	Weight Lbs.
3	C-715	1/2	2	12.63	4.88	1.34	2	1.38	7.81
5	D-715	1/2, 5/8	2.5	15.44	6.5	1.81	2.5	1.81	14.87
8.5	EE-715	1/2, 5/8, 3/4	2.88	18.56	8.81	2.25	3.25	2.59	29.25
10	G-715	3/4, 7/8, 1	4	23.63	11	3	3.25	3	60
15	GG-715	3/4, 7/8, 1	4	23.63	11	3	4.25	3	60
25	H-715	1 1/8, 1 1/4	5	32	12.02	3.54	4.41	4.21	145
35	HH-715	1 1/8, 1 1/4	5	32	12.02	3.54	4.41	4.21	145

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- ECONO-LINK, THRUST BEARING SWIVELER



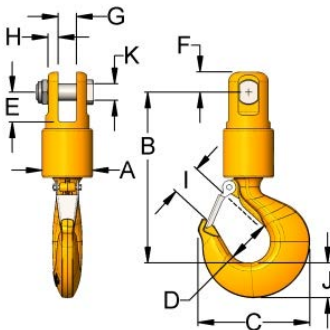
- Single tapered thrust bearing, factory lubricated
- Tight-fit bronze bushing acting as a seal, retains lubricant and protects bearing
- Load capacities from 3 to 15 tons
- Wire rope diameters from ½" to 1"
- Six popular clevis, hook, and eye configurations available
- Structural components from cast alloy steel
- Hook models with forged alloy steel hook and safety latch
- Finished with enamel coating
- 4:1 design factor, Proof Load is 2 x WLL

NEW! Insulated
*Insulated Swivel for Suspended
 Welding & Other DC current
 protection applications*



- Includes all the same, proven Econo-Link features, plus advanced electrical protection
- Rated for 2000 volts direct current (DC only)
- Electrical test certificates available
- To order, add suffix "-DC", e.g., 3E185-DC

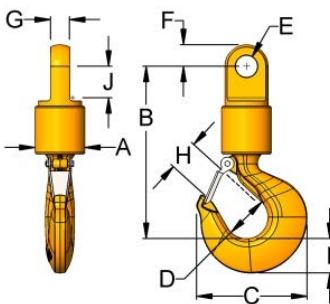
CLEVIS TO HOOK - TYPE 185



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	Weight Lbs.
3	3E185	1/2	2.63	8.12	4.84	2.00	1.38	1.19	1.00	.50	1.36	1.44	.75	8.8
6	6E185	5/8	3.34	10.24	6.28	2.50	1.94	1.31	1.13	.75	1.61	1.82	1.00	18.8
10	10E185	3/4	3.75	12.78	8.34	3.25	2.25	1.50	1.38	.75	2.27	2.60	1.25	34.9
15	15E185	1	4.13	15.77	10.34	4.25	2.56	1.81	1.75	.88	3.02	3.01	1.38	56.0

* Design factor 4:1

EYE TO HOOK - TYPE 285

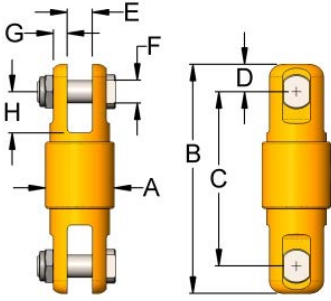


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	Weight Lbs.
3	3E285	1/2	2.63	7.87	4.84	2.00	1.06	1.13	1.00	1.36	1.44	1.13	8.1
6	6E285	5/8	3.34	9.75	6.28	2.50	1.41	1.50	1.19	1.61	1.82	1.38	16.7
10	10E285	3/4	3.75	13.03	8.34	3.25	1.66	1.63	1.44	2.27	2.60	2.38	33.8
15	15E285	1	4.13	15.95	10.34	4.25	2.06	1.88	1.94	3.02	3.01	2.63	55.8

* Design factor 4:1

SWIVELS- ECONO-LINK, THRUST BEARING SWIVELER

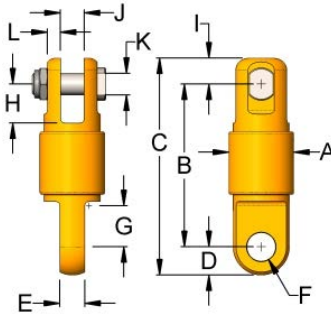
CLEVIS TO CLEVIS - TYPE 181



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
3	3E181	1/2	2.63	6.27	8.65	1.19	1.00	.75	0.5	1.38	7.7
6	6E181	5/8	3.34	8.16	10.78	1.31	1.13	1.00	0.75	1.94	16.7
10	10E181	3/4	3.75	9.53	12.53	1.50	1.38	1.25	0.75	2.25	23.7
15	15E181	1	4.13	11.19	14.81	1.81	1.75	1.38	0.88	2.56	33.8

* Design factor 4:1

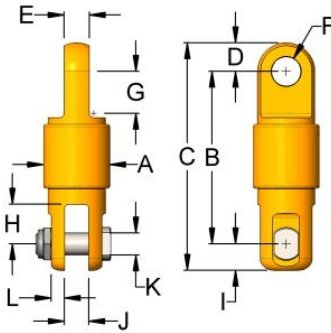
CLEVIS TO EYE - TYPE 182



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
3	3E182	1/2	2.63	5.90	8.21	1.13	1.00	1.06	1.13	1.38	1.19	1.00	.75	.50	6.9
6	6E182	5/8	3.34	7.41	10.22	1.5	1.19	1.41	1.38	1.94	1.31	1.13	1.00	0.75	14.2
10	10E182	3/4	3.75	9.41	12.53	1.63	1.44	1.66	2.38	2.25	1.50	1.38	1.25	0.75	21.6
15	15E182	1	4.13	11.13	14.81	1.88	1.94	2.06	2.63	2.56	1.81	1.75	1.38	0.88	31.9

* Design factor 4:1

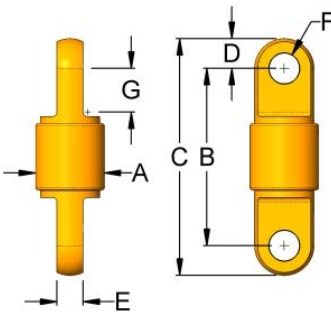
EYE TO CLEVIS - TYPE 281



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
3	3E281	1/2	2.63	6.02	8.34	1.13	1.00	1.06	1.13	1.38	1.19	1.00	.75	.50	7.1
6	6E281	5/8	3.34	7.66	10.47	1.50	1.19	1.41	1.38	1.94	1.31	1.13	1.00	0.75	14.8
10	10E281	3/4	3.75	9.78	12.91	1.63	1.44	1.66	2.38	2.25	1.50	1.38	1.25	0.75	22.6
15	15E281	1	4.13	11.38	15.06	1.88	1.94	2.06	2.63	2.56	1.81	1.75	1.38	0.88	33.6

* Design factor 4:1

EYE TO EYE - TYPE 282



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
3	3E282	1/2	2.63	5.65	7.90	1.13	1.00	1.06	1.13	6.3
6	6E282	5/8	3.34	6.91	9.91	1.50	1.19	1.41	1.38	12.4
10	10E282	3/4	3.75	9.66	12.91	1.63	1.44	1.66	2.38	20.6
15	15E282	1	4.13	11.31	15.06	1.88	1.94	2.06	2.63	31.8

* Design factor 4:1

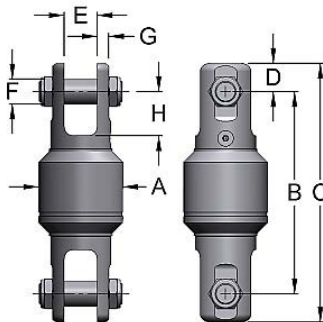
SWIVELS- HD-LINK, HEAVY DUTY (ANGULAR CONTACT)



- Heavy Duty (HD) barrel for maximized swivel life in rough conditions
- Thick-walled barrel protects bearings and seals from shock and dust
- Equipped with low torque angular contact ball bearings like Miller Y-Link
- Designed for high impact, abrasive environments as found in demolition, quarries, etc.
- Machined from solid forged alloy steel bar
- Zinc plating inside and out for effective protection against corrosion
- Grease fitting allows for easy, frequent lubrication
- For Working Loads from 5 to 35 tons and wire rope sizes 5/8" to 1 1/2"
- Higher capacities are available upon request

For demolition, the swivel selected should be one size larger than the normally recommended size, should be lubricated daily, and should be installed 4-6 feet above the breaking ball overhaul weight.

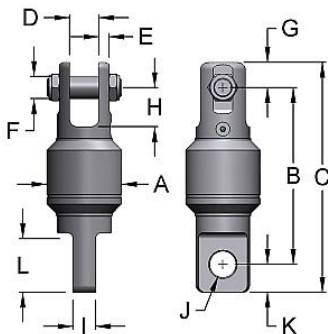
CLEVIS TO CLEVIS - TYPE 131



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
5		5/8	3.5	7.94	10.19	1.13	1	.88	0.56	1.56	12
5	D-131	5/8	3.5	7.94	10.19	1.13	1	.88	0.56	1.56	12
8.5	EE-131	3/4	4	9.63	12.31	1.34	1.56	1.19	0.56	2.09	19.75
15	GG-131	1	5	14	17.5	1.75	1.75	1.5	0.81	3.5	50
35	HH-131	1 1/2	6	15.94	20.69	2.38	2	2	1.13	3.69	99
45	I-131		7	20.06	26.06	3	2.5	2.25	1.25	4	160

* Design factors for WLL=35T and below are 5:1; and 45T are 4:1

CLEVIS TO EYE - TYPE 132



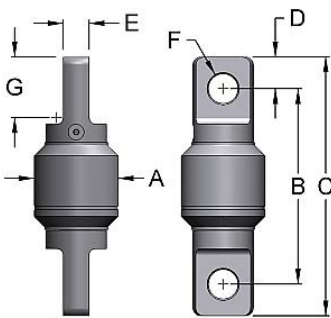
WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
5	D-132	5/8	3.5	7.88	10.19	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	11.38
8.5	EE-132	3/4	4	9.5	12.34	1.56	0.56	1.19	1.34	2.09	1.25	1.41	1.5	3.13	19.75
10	G-132	7/8	5	13.94	17.5	1.75	0.81	1.5	1.75	3.5	1.72	1.66	1.81	4.66	49
15	GG-132	1	5	13.63	17.5	1.75	0.81	1.5	1.75	3.5	2	2.03	2.13	4.91	50
35	HH-132	1 1/2	6	16.44	21.19	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	96
45	I-132		7	20.06	26.06	2.5	1.25	2.25	3	4	2.5	2.53	3	7	155

* Design factors for WLL=35T and below are 5:1; and 45T are 4:1



SWIVELS- HD-LINK, HEAVY DUTY (ANGULAR CONTACT)

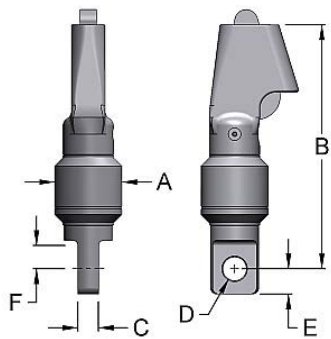
EYE TO EYE - TYPE 232



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
5	D-232	5/8	3.5	7.69	10.19	1.19	1	1.28	2.5	11
8.5	EE-232	3/4	4	9.31	12.34	1.5	1.25	1.41	3.13	19.25
10	G-232	7/8	5	13.88	17.5	1.81	1.72	1.66	4.66	47
15	GG-232	1	5	13.25	17.5	2.13	2	2.03	4.91	48
35	HH-232	1.5	6	16.5	21.19	2.38	2.25	2.31	5.25	93
45	I-232		7	20.06	26.06	3	2.5	2.53	7	150

* Design factors for WLL=35T and below are 5:1; and 45T are 4:1

WEDGE TO EYE - TYPE 732



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	Weight Lbs.
5	D-732	1/2, 5/8	3.5	12.5	1	1.28	1.19	1.31	15
8.5	EE-732	1/2, 5/8, 3/4	4	14.25	1.25	1.41	1.5	1.63	25
10	G-732	3/4, 7/8, 1	5	19.69	1.72	1.66	1.81	2.84	55
15	GG-732	3/4, 7/8, 1	5	19.38	2	2.03	2.13	2.78	56
25	H-732	1 1/8, 1 1/4	6	25.44	2.25	2.31	2.38	2.88	119
35	HH-732	1 1/8, 1 1/4	6	25.44	2.25	2.31	2.38	2.88	119

* Design factor 5:1

Load Handling SOLUTIONS

- ▶ Swivels
- ▶ Forged Hooks
- ▶ Blocks
- ▶ Sheaves

SCR Pull-in Swivel
375 metric tons working load

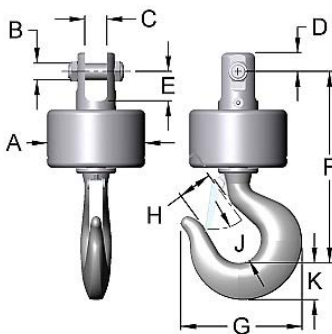
www.millerproducts.net

SWIVELS- g-LINK, OVERHAUL (ANGULAR CONTACT)



- Miller's **g-Link** overhaul swivels weigh 2 ½ to nearly four times as much standard swivels of the same type
- Adds overhaul weight without increasing overall length
- Ideal for small truck cranes and low head room applications where only minimal overhaul weight is required
- For 1.5 to 8.5 ton Working Loads
- Angular contact ball bearings for low torque swivel motion
- Machined from forged alloy steel bar
- Zinc plating inside and out for effective protection against corrosion.
- The g-Link is available with many of the other end fittings shown in this catalog. If you do not see your requirement in the following tables, please inquire.

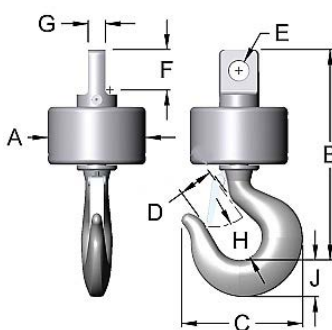
CLEVIS TO HOOK - TYPE 125



WLL* Tons	Model Number	Rope size	A	B	C	D	E	F	G	H	J	K	Weight Lbs.
1.5	B-125	3/8	4	0.5	0.5	.69	0.78	6.09	4.09	1.13	1.19	1.13	9
3	C-125	1/2	5	0.75	0.75	.94	1.19	8.38	4.94	1.34	1.38	1.48	17.94
5	D-125	5/8	7	0.88	1	1.13	1.56	12.63	6.5	1.69	1.75	1.81	43.81
8.5	EE-125	3/4	7	1.19	1.56	1.34	2.09	13.75	8.94	2.25	2.56	2.59	60

* Design factor 5:1 * Models 5 Ton and up include lubrication fitting.

EYE TO HOOK - TYPE 225



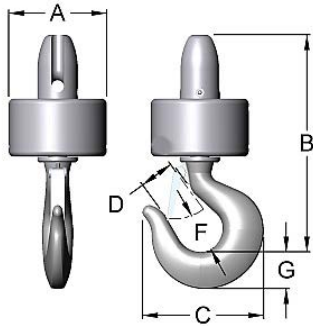
WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	J	Weight Lbs.
1.5	B-225	3/8	4	6.25	4.09	1.13	0.66	1.38	0.5	1.19	1.13	9
3	C-225	1/2	5	9.75	4.81	1.34	0.91	1.88	0.75	1.38	1.44	17.94
5	D-225	5/8	7	12.5	6.5	1.69	1.28	2.5	1	1.75	1.81	43.81
8.5	EE-225	3/4	7	13.5	8.69	2.25	1.41	3.13	1.25	2.56	2.59	60

* Design factor 5:1 * Models 5 Ton and up include lubrication fitting.



SWIVELS- g-LINK, OVERHAUL (ANGULAR CONTACT)

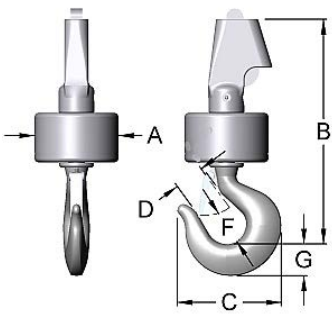
THIMBLE TO HOOK - TYPE 625



WLL* Tons	Model Number	Rope Size	A	B	C	D	F	G	Weight Lbs.
1.5	B-625	3/8	4	8.22	4.09	1.13	1.19	1.13	9.38
3	C-625	1/2	5	10.5	4.94	1.34	1.38	1.19	19.19
5	D-625	5/8	7	14.22	6.5	1.69	1.75	1.81	45.31
8.5	EE-625	3/4	7	15.5	8.69	2.25	2.19	2.59	56.25

* Design factor 5:1 * Models 5 Ton and up include lubrication fitting.

WEDGE TO HOOK - TYPE 725



WLL* Tons	Model Number	Rope Size	A	B	C	D	F	G	Weight Lbs.
3	C-725	1/2	5	12.69	4.94	1.34	1.13	1.19	20.38
5	D-725	1/2, 5/8	7	17.19	6.5	1.69	1.75	1.81	47.88
8.5	EE-725	1/2, 5/8, 3/4	7	18.5	8.69	2.25	2.56	2.59	65.25

* Design factor 5:1 * Models 5 Ton and up include lubrication fitting.

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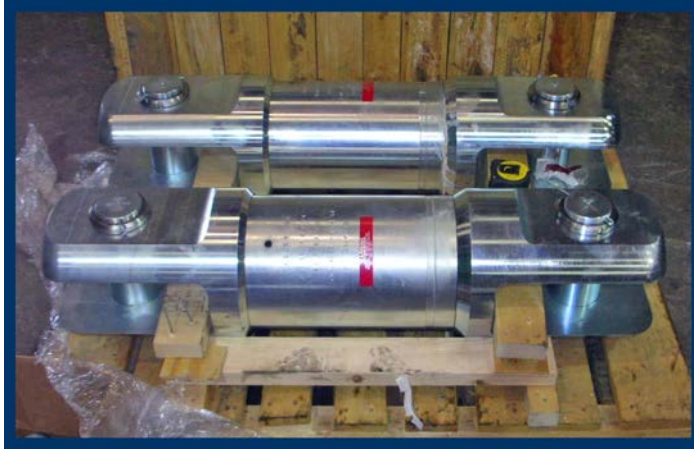
Load Handling SOLUTIONS

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CELEBRATING OUR 80TH YEAR!

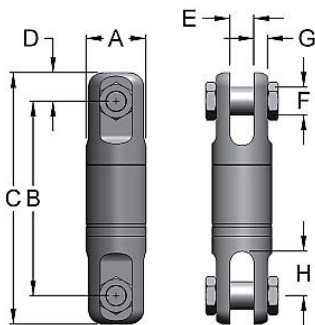
SWIVELS- Y-LINK/HP, HIGH PRESSURE



Miller Y-Link alloy steel swivels are available in a high pressure version in six popular configurations, suitable for use in submerged environments where corrosion exposure from long term unattended deployments are not a major concern. These swivels are based on our Y-Link models and contain special pressure compensating seals for reliable operation to ocean depths of 10,000 feet. Y-Link/HP models are slightly longer than standard Y-Link models due to the addition of special high pressure seals.

- **Y-Link/HP** high pressure swivels contain special pressure compensating seals and oil-filled cavities for reliable operation to depths of 10,000 feet
- Working Load Limits from 0.75 to 600 tons
- Angular contact bearing design extends the life of the wire rope, and maximizes efficiency, reliability and service life
- Alloy steel structural components are zinc plated inside and out for corrosion protection
- Design Factor of 5 to 1 minimum for swivels 35 ton and below
- Design Factor of 4 to 1 minimum for swivels 45 tons and above
- Proof Load is 2 x WLL
- Models 8.5 tons and up include lubrication fitting
- Special designs are available

CLEVIS TO CLEVIS - TYPE 111HP

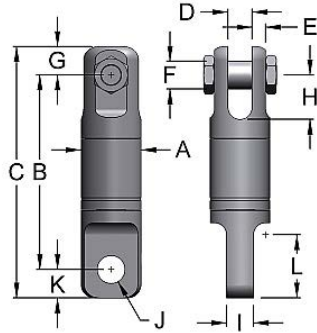


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
0.75	BB-111HP	.25	1.31	3.69	5.56	.44	0.31	.38	0.22	0.56	1
1.5	B-111HP	.375	1.63	5	6.38	.69	0.5	.5	0.31	0.78	2
3	C-111HP	.5	2	5.44	9.31	.94	0.75	.75	0.38	1.19	5
5	D-111HP	.625	2.5	9.19	11.44	1.13	1	.88	0.56	1.56	10
8.5	EE-111HP	.75	3	10.38	13.06	1.34	1.56	1.19	0.53	2.03	15
15	GG-111HP	1	4	15.5	19	1.75	1.75	1.5	0.81	3.5	43
35	HH-111HP	1.5	5	18.19	22.94	2.38	2	2	1.13	3.69	85
45	I-111HP		6	22.56	28.56	3	2.5	2.25	1.25	4	156
60	JJ-111HP		7	24	31	3.5	3	2.5	1.5	4.5	229
75	K-111HP		8	27	35	4	3.5	3	1.75	5	328
100	KK-111HP		10	36.5	47.5	5.5	4.06	4	2	6.75	747
200	L-111HP		11.5	36.52	48.02	5.75	5.06	4.5	2.56	7.75	985
300	M-111HP		14	44.2	59.2	7.5	6.13	4.75	3.94	9.5	1680

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK/HP, HIGH PRESSURE

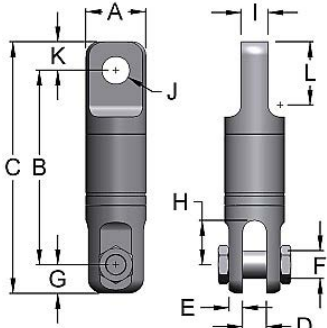
CLEVIS TO EYE - TYPE 112HP



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-112HP	.25	1.31	4.75	5.56	.31	0.22	.38	0.44	0.56	.31	.38	.44	0.94	1
1.5	B-112HP	.375	1.63	5	6.31	0.5	0.31	.5	0.68	0.78	0.5	.66	.63	1.34	2
3	C-112HP	.5	2	7.31	9.31	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	5
5	D-112HP	.625	2.5	9.13	11.44	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	11
8.5	EE-112HP	.75	3	10.25	13.09	1.56	0.53	1.19	1.34	2.09	1.25	1.41	1.5	3.13	16
15	GG-112HP	1	4	15.25	19.13	1.75	0.81	1.5	1.75	3.5	2	2.03	2.13	4.91	43
35	HH-112HP	1.5	5	18.19	22.94	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	82
45	I-112HP		6	22.56	28.56	2.5	1.25	2.25	3	4	2.5	2.53	3	7	149
60	JJ-112HP		7	24	31	3	1.5	2.5	3.5	4.5	3.5	2.88	3.5	7	226
75	K-112HP		8	27	35	3.5	1.75	3	4	5	3.5	3.38	4	8.13	333
100	KK-112HP		10	36.5	47.5	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	698
200	L-112HP		11.5	35.52	47.27	5.06	2.56	4.5	5.75	7.75	5	4.81	6	13	925
300	M-112HP		14	44.2	59.2	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

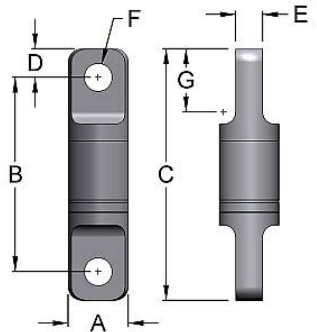
EYE TO CLEVIS - TYPE 211HP



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-211HP	1/4	1.31	4.69	5.56	.31	0.22	.38	0.44	0.56	.31	.38	.44	0.94	1
1.5	B-211HP	3/8	1.63	5	6.31	0.5	0.31	.5	0.68	0.78	0.5	.66	.63	1.34	2
3	C-211HP	1/2	2	7.31	9.31	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	5
5	D-211HP	5/8	2.5	9.13	11.44	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	10
8.5	EE-211HP	3/4	3	10.25	13.09	1.56	0.53	1.19	1.34	2.09	1.25	1.41	1.5	3.13	16
15	GG-211HP	1	4	15.25	19.13	1.75	0.81	1.5	1.75	3.5	2	2.03	2.13	4.91	43
35	HH-211HP	1 1/2	5	18.19	22.94	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	80
45	I-211HP		6	22.56	28.56	2.5	1.25	2.25	3	4	2.5	2.53	3	7	149
60	JJ-211HP		7	24	31	3	1.5	2.5	3.5	4.5	3.5	2.88	3.5	7	233
75	K-211HP		8	27	35	3.5	1.75	3	4	5	3.5	3.38	4	8.25	333
100	KK-211HP		10	36.5	47.5	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	731
200	L-211HP		11.5	36.67	48.42	5.06	2.56	4.5	5.75	7.75	5	4.81	6	13	960
300	M-211HP		14	44.2	59.2	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

EYE TO EYE - TYPE 212HP

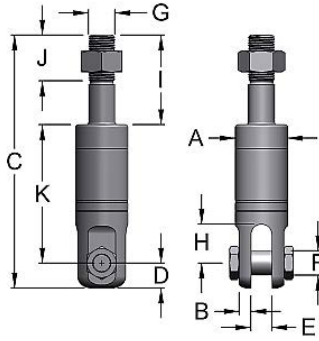


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-212HP	1/4	1.31	4.94	5.88	.44	0.31	.39	0.94	1
1.5	B-212HP	3/8	1.63	5	6.25	.69	0.5	.66	1.34	2
3	C-212HP	1/2	2	7.31	9.31	1	0.75	.91	1.94	5
5	D-212HP	5/8	2.5	9.06	11.44	1.19	1	1.28	2.5	10
8.5	EE-212HP	3/4	3	10.13	13.06	1.5	1.25	1.41	3.13	16
15	GG-212HP	1	4	15.13	19.38	2.13	2	2.03	4.91	43
35	HH-212HP	1 1/2	5	18.19	22.94	2.38	2.25	2.31	5.25	79
45	I-212HP		6	22.56	28.56	3	2.5	2.53	7	142
60	JJ-212HP		7	24	31	3.5	3.5	2.88	7	231
75	K-212HP		8	27	35	4	3.5	3.38	8.25	340
100	KK-212HP		10	36.5	47.5	5.5	4.5	4.03	11.38	704
200	L-212HP		11.5	35.67	47.67	6	5	4.81	13	900
300	M-212HP		14	43.2	59.2	8	6	6.06	17	1600

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- Y-LINK/HP, HIGH PRESSURE

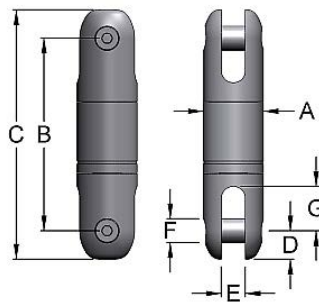
STUD TO CLEVIS - TYPE 311HP



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	Weight Lbs.
0.75	BB-311HP	1/4	1.31	0.22	5.81	.44	0.31	.38	0.63	0.56	2	1.5	4.38	1
1.5	B-311HP	3/8	1.63	0.31	7	.69	0.5	.75	0.75	0.78	2	1.5	4.31	2
3	C-311HP	1/2	2	0.38	10.44	.88	0.75	.75	0.88	1.19	3	2.63	5.25	5
5	D-311HP	5/8	2.5	0.56	11.19	1.13	1	.88	1.25	1.56	3.5	3.13	6.56	11
8.5	EE-311HP	3/4	3	0.53	13.78	1.34	1.56	1.19	1.5	2.09	4	3	8.44	15
15	GG-311HP	1	4	0.81	18.69	1.75	1.75	1.5	2	3.5	5.5	3	11.44	40
35	HH-311HP	1 1/2	5	1.13	25.56	2.38	2	2	2.5	3.69	8.5	4.25	14.69	84
45	I-311HP		6	1.25	28	3	2.5	2.25	3	4	8	4	17	141
60	JJ-311HP		7	1.5	31.5	3.5	3	2.5	3.25	4.5	8	4	20	219
75	K-311HP		8	1.75	36.5	4	3.5	3	3.75	5	10	5	22.5	295
100	KK-311HP		10	2	54.75	5.5	4.06	4	6.5	6.75	20.5	7.25	28.75	699

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

BULLET NOSE TO BULLET NOSE - TYPE 414HP

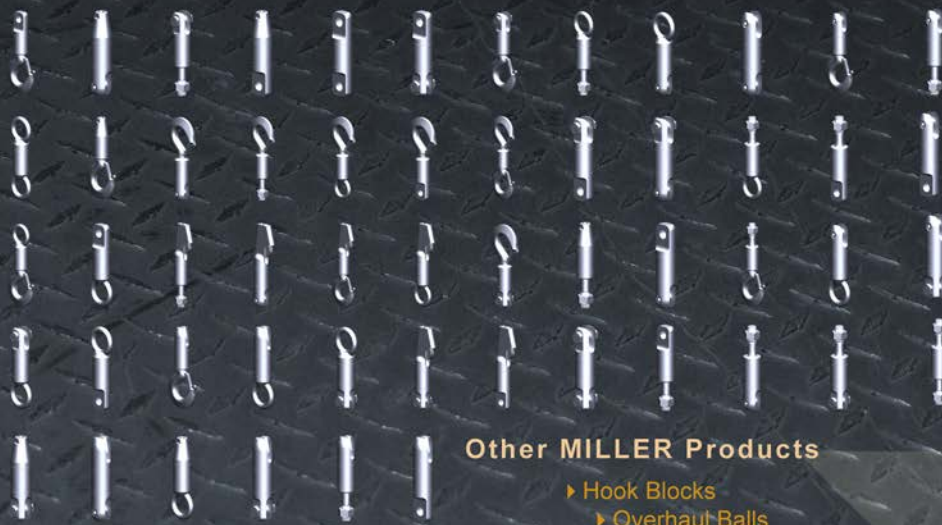


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-414HP	.25	1.31	4.88	5.63	.44	0.31	.38	0.56	1
1.5	B-414HP	3/8	1.63	5	6.25	.56	0.5	.44	0.81	2
3	C-414HP	3/8	2	6.5	8.13	.81	0.75	.63	0.94	4
5	D-414HP	5/8	2.5	9.25	11.38	1.13	1	.88	1.56	9
8.5	EE-414HP	3/4	3	10.56	13.13	1.25	1.28	1	2.13	16
15	GG-414HP	1	4	15	18.5	1.75	1.75	1.5	3.25	42
35	HH-414HP	1 1/2	5	18.16	22.91	2.38	2	2	3.69	87

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

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SWIVELS- HYDRO-LINK & HYDRO-LINK/HP

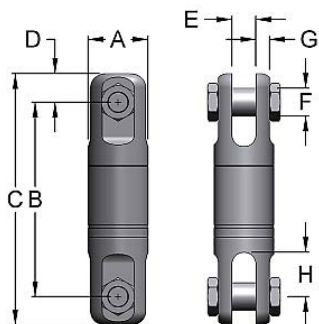
- Miller Hydro-Link swivels are machined from a special non-magnetic, austenitic stainless steel alloy of exceptionally high strength and corrosion resistance which maintains good mechanical properties at elevated and sub-zero temperatures.
- **Available in two configurations, standard Hydro-Link and high pressure Hydro-Link / HP**



- **Hydro-Link** swivels are intended for use with cable and chain in non-submerged corrosive environments such as sea air or harsh industrial environments
- **Hydro-Link / HP** high pressure swivels contain special pressure compensating seals and oil-filled cavities for reliable operation to depths of 10,000 feet
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- Angular contact bearing design extends the life of the wire rope, and maximizes efficiency, reliability and service life
- Stainless steel parts are passivated for added corrosion resistance
- Design Factor of 5 to 1 minimum for swivels 35 tons and below
- Design Factor of 4 to 1 minimum for swivels 45 tons and above
- Proof Load is 2 x WLL
- Models 8.5 tons and up include lubrication fitting
- Special designs available

SWIVELS- HYDRO LINK, STAINLESS STEEL

CLEVIS TO CLEVIS - TYPE 151, STAINLESS STEEL

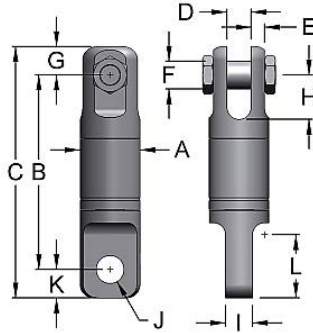


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	H	G	Weight Lbs.
0.75	BB-151	1/4	1.31	3.5	4.38	0.44	0.31	.38	.56	0.22	0.93
1.5	B-151	3/8	1.63	4.06	5.44	0.69	0.5	.5	.78	0.31	2
3	C-151	1/2	2.00	6.25	8.09	0.94	0.75	.75	1.19	0.38	4.5
5	D-151	5/8	2.5	7.94	10.09	1.13	1	.88	1.56	0.56	9.5
8.5	EE-151	3/4	2.88	9.63	12.31	1.34	1.56	1.19	2.09	0.53	15.25
10	G-151	7/8	4.00	14	17.5	1.75	1.75	1.5	3.5	0.81	40
15	GG-151	1	4.00	14	17.5	1.75	1.75	1.5	3.5	0.81	40
25	H-151	1 1/4	5.00	15.94	20.69	2.38	2	2	3.69	1.13	78
35	HH-151	1 1/2	5.00	15.94	20.69	2.38	2	2	3.69	1.13	78
45	I-151		6.00	20.06	26.06	3	2.5	2.25	4	1.25	146
50	J-151		7.00	20.94	27.94	3.5	3	2.5	4.5	1.5	207
60	JJ-151		7.00	20.94	27.94	3.5	3	2.5	4.5	1.5	207
75	K-151		8.00	24	32	4	3.5	3	5	1.75	300
100	KK-151		10.00	32.38	43.38	5.5	4.06	4	6.75	2	692
150	KKK-151		10	32.88	43.88	5.5	4.06	4	6.75	2	692
200	L-151		11.25	36.31	48.31	6	5.06	4.5	8.25	3.09	885
300	M-151		14	43.95	58.95	7.5	6.13	4.75	9.5	3.94	1680

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- HYDRO LINK, STAINLESS STEEL

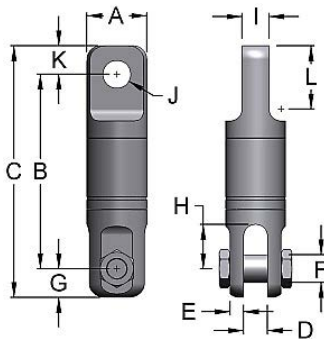
CLEVIS TO EYE - TYPE 152, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-152	1/4	1.31	3.56	4.44	0.31	0.22	.38	0.44	0.56	0.31	.38	.44	0.94	0.94
1.5	B-152	3/8	1.63	4.13	5.38	0.5	0.31	.5	0.69	0.78	0.5	.66	.63	1.34	1.94
3	C-152	1/2	2.0	6.19	8.13	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	4.25
5	D-152	5/8	2.5	7.88	10.19	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	8.94
8.5	EE-152	3/4	2.88	9.5	12.34	1.56	0.53	1.19	1.34	2.19	1.25	1.41	1.5	3.13	15.25
10	G-152	7/8	4.0	13.94	17.5	1.75	0.81	1.5	1.75	3.5	1.72	1.66	1.81	4.66	39
15	GG-152	1	4.0	13.63	17.5	1.75	0.81	1.5	1.75	3.5	1.94	2.03	2.13	4.91	40
25	H-152	1 1/4	5.0	15.94	20.69	2	1.13	2.03	2.38	3.69	2.25	2.31	2.38	5.25	75
35	HH-152	1 1/2	5.0	15.94	20.69	2	1.13	2.03	2.38	3.69	2.25	2.31	2.38	5.25	75
45	I-152		6.0	20.06	26.06	2.5	1.25	2.25	3	4	2.5	2.53	3	7	139
50	J-152		7.0	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	205
60	JJ-152		7.0	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	205
75	K-152		8.0	24	32	3.5	1.75	3	4	5	3.5	3.38	4	8.13	305
100	KK-152		10.0	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	647
150	KKK-152		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.28	5.5	11.38	647
200	L-152		11.25	35.92	48.29	5.06	3.09	4.5	6	8.25	5	4.81	6.38	13.5	885
300	M-152		14	43.45	58.95	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

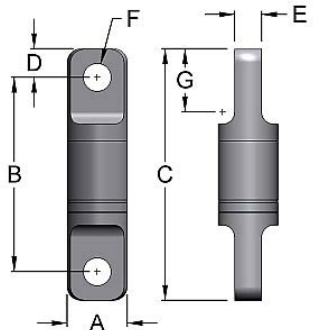
EYE TO CLEVIS - TYPE 251, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-251	1/4	1.31	3.63	4.5	0.31	0.22	.38	0.44	0.44	0.31	.39	.44	0.94	0.94
1.5	B-251	3/8	1.63	4.13	5.44	0.5	0.31	.5	0.69	0.78	0.5	.66	.63	1.34	1.88
3	C-251	1/2	2.0	6.19	8.13	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	4.31
5	D-251	5/8	2.5	7.88	10.19	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	9.06
8.5	EE-251	3/4	2.88	9.5	12.34	1.56	0.53	1.19	1.34	2.19	1.25	1.41	1.5	3.13	14.75
10	G-251	7/8	4.0	13.94	17.5	1.75	0.81	1.5	1.75	3.5	1.72	1.66	1.81	4.66	39
15	GG-251	1	4.0	13.63	17.5	1.75	0.81	1.5	1.75	3.5	1.94	2.03	2.13	4.91	40
25	H-251	1 1/4	5.0	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
35	HH-251	1 1/2	5.0	15.94	20.69	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	75
45	I-251		6.0	20.06	26.06	2.5	1.25	2.25	3	4	2.5	2.53	3	7	139
50	J-251		7.0	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	211
60	JJ-251		7.0	20.94	27.94	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	211
75	K-251		8.0	24	32	3.5	1.75	3	4	5	3.5	3.38	4	8.13	305
100	KK-251		10.0	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	678
150	KKK-251		10	32.88	43.88	4.06	2	4	5.5	6.75	4.5	4.28	5.5	11.38	678
200	L-251		11.25	35.93	48.31	5.06	3.09	4.5	6	8.25	5	4.81	6.38	13.5	885
300	M-251		14	43.45	58.95	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

EYE TO EYE - TYPE 252, STAINLESS STEEL



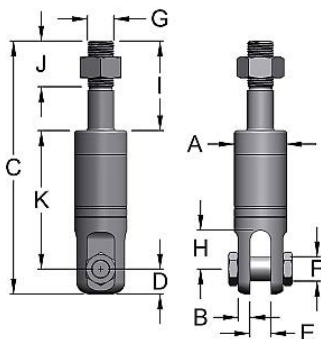
WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-252	1/4	1.31	3.81	4.69	0.44	0.31	.39	0.94	0.93
1.5	B-252	3/8	1.63	4.13	5.38	0.63	0.5	.66	1.34	1.87
3	C-252	1/2	2	6.13	8.13	1	0.75	.91	1.94	4.06
5	D-252	5/8	2.5	7.81	10.19	1.19	1	1.28	2.5	8.5
8.5	EE-252	3/4	2.88	9.31	12.31	1.5	1.25	1.41	3.13	14.75
10	G-252	7/8	4	13.88	17.5	1.81	1.72	1.63	4.66	37
15	GG-252	1	4	13.25	17.5	2.13	1.94	2.03	4.91	39
25	H-252	1 1/4	5	15.94	20.69	2.38	2.25	2.31	5.25	72
35	HH-252	1 1/2	5	15.94	20.69	2.38	2.25	2.31	5.25	72
45	I-252		6	20.06	26.06	3	2.5	2.53	7	132
50	J-252		7	20.94	27.94	3.5	2.94	2.88	7	209
60	JJ-252		7	20.94	27.94	3.5	2.94	2.88	7	209
75	K-252		8	24	32	4	3.5	3.38	8.13	311
100	KK-252		10	32.88	43.41	5.5	4.5	4.03	11.38	653
150	KKK-252		10	32.88	43.88	5.5	4.5	4.28	11.38	657
200	L-252		11.25	35.56	48.31	6.38	5	4.81	13.5	880
300	M-252		14	42.95	58.95	8	6	6.06	17	1610

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1



SWIVELS- HYDRO LINK, STAINLESS STEEL

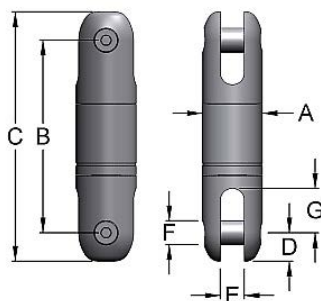
STUD TO CLEVIS - TYPE 351, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	Weight Lbs.
0.75	BB-351	1/4	1.31	0.22	5.63	0.44	0.31	.38	0.63	0.56	2	1.5	3.19	1.06
1.5	B-351	3/8	1.63	0.31	6.06	0.69	0.5	.5	0.75	0.78	2	1.5	3.75	1.87
3	C-351	1/2	2	0.38	9.19	0.94	0.75	.75	0.88	1.19	3	2.63	5.25	4.5
5	D-351	5/8	2.5	0.56	11.19	1.13	1	.88	1.25	1.56	3.5	3.13	6.56	9.75
8.5	EE-351	3/4	2.88	0.53	13.03	1.56	1.56	1.19	1.5	2.19	4	3	7.69	14.5
10	G-351	7/8	4	0.81	17.44	1.75	1.75	1.5	2	3.5	5.5	3	10.19	37
15	GG-351	1	4	0.81	17.44	1.75	1.75	1.5	2	3.5	5.5	3	10.19	37
25	H-351	1 1/4	5	1.13	23.31	2.38	2	2	2.5	3.69	8.5	4.25	12.44	78
35	HH-351	1 1/2	5	1.13	23.31	2.38	2	2	2.5	3.69	8.5	4.25	10.44	78
45	I-351		6	1.25	25.56	3	2.5	2.25	3	4	8	4	14.56	131
50	J-351		7	1.5	28.44	3.5	3	2.5	3.25	4.5	8	4	16.94	198
60	JJ-351		7	1.5	28.44	3.5	3	2.5	3.25	4.5	8	4	16.94	198
75	K-351		8	1.75	33.5	4	3.5	3	3.75	5	10	5	19.5	271
100	KK-351		10	2	51.13	5.5	4.06	4	6.5	6.75	20.5	7.25	25.13	648

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

BULLET NOSE TO BULLET NOSE - TYPE 454, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-454	1/4	1.31	3.56	4.44	0.44	0.31	.38	0.56	1.12
1.5	B-454	3/8	1.63	4.06	5.19	0.56	0.5	.44	0.81	1.81
3	C-454	1/2	2	5.44	7.06	0.81	0.75	.63	0.94	3.87
5	D-454	5/8	2.5	7.88	10.19	1.13	1	.88	1.56	8.06
8.5	EE-454	3/4	2.88	9.81	12.31	1.25	1.28	1	2.13	14.5
10	G-454	7/8	4	13.25	16.75	1.75	1.75	1.5	3.25	40
15	GG-454	1	4	13.25	16.75	1.75	1.75	1.5	3.25	40
25	H-454	1 1/4	5	15.91	20.66	2.38	2	2	3.69	84
35	HH-454	1 1/2	5	15.91	20.66	2.38	2	2	3.69	84

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

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Y-Link, Angular Contact Ball Bearing

Econo-Link, Tapered Roller Thrust Bearing

Overhaul Ball Assemblies

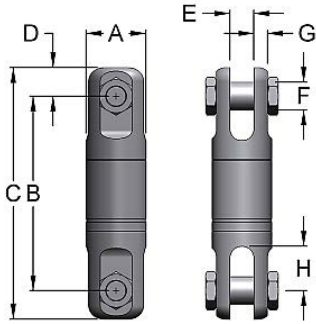
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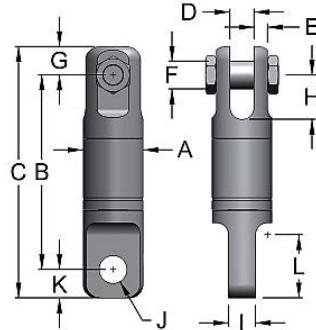
CLEVIS TO CLEVIS - TYPE 151HP, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	Weight Lbs.
0.75	BB-151HP	1/4	1.31	3.69	5.56	.44	0.31	.38	0.22	0.56	1
1.5	B-151HP	3/8	1.63	5	6.38	.69	0.5	.5	0.31	0.78	2
3	C-151HP	1/2	2	5.44	9.31	.94	0.75	.75	0.38	1.19	5
5	D-151HP	5/8	2.5	9.19	11.44	1.13	1	.88	0.56	1.56	10
8.5	EE-151HP	3/4	3	10.38	13.06	1.34	1.56	1.19	0.53	2.03	15
15	GG-151HP	1	4	15.5	19	1.75	1.75	1.5	0.81	3.5	43
35	HH-151HP	1 1/2	5	18.19	22.94	2.38	2	2	1.13	3.69	85
45	I-151HP		6	22.56	28.56	3	2.5	2.25	1.25	4	156
60	JJ-151HP		7	24	31	3.5	3	2.5	1.5	4.5	229
75	K-151HP		8	27	35	4	3.5	3	1.75	5	328
100	KK-151HP		10	36.5	47.5	5.5	4.06	4	2	6.75	747
200	L-151HP		11.25	36.31	48.31	6	5.06	4.5	3.09	8.25	885
300	M-151HP		14	44.2	59.2	7.5	6.13	4.75	3.94	9.5	1680

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

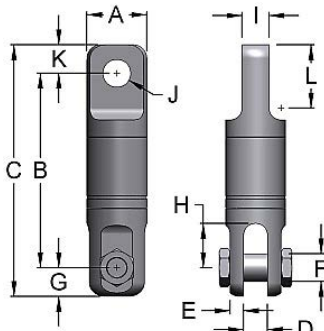
CLEVIS TO EYE - TYPE 152HP, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-152HP	1/4	1.31	4.75	5.56	.31	0.22	.375	0.44	0.56	.31	.38	.44	0.94	1
1.5	B-152HP	3/8	1.63	5	6.31	0.5	0.31	.5	0.68	0.78	0.5	.66	.63	1.34	2
3	C-152HP	1/2	2	7.31	9.31	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	5
5	D-152HP	5/8	2.5	9.13	11.44	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	11
8.5	EE-152HP	3/4	3	10.25	13.09	1.56	0.53	1.06	1.34	2.09	1.25	1.41	1.5	3.13	16
15	GG-152HP	1	4	15.25	19.13	1.75	0.81	1.5	1.75	3.5	2	2.03	2.13	4.91	43
35	HH-152HP	1 1/2	5	18.19	22.94	2	1.13	2.03	2.38	3.69	2.25	2.31	2.38	5.25	82
45	I-152HP		6	22.56	28.56	2.5	1.25	2.25	3	4	2.5	2.53	3	7	149
60	JJ-152HP		7	24	31	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	226
75	K-152HP		8	27	35	3.5	1.75	3	4	5	3.5	3.38	4	8.13	333
100	KK-152HP		10	36.5	47.5	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	698
200	L-152HP		11.25	35.92	48.29	5.06	3.09	4.5	6	8.25	5	4.81	6.38	13.5	885
300	M-152HP		14	44.2	59.2	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

EYE TO CLEVIS - TYPE 251HP, STAINLESS STEEL

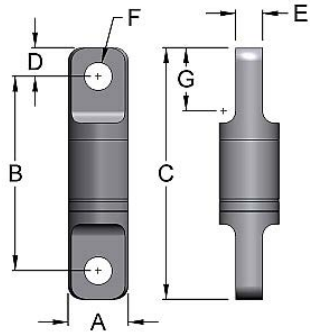


WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	L	Weight Lbs.
0.75	BB-251HP	1/4	1.31	4.69	5.56	.31	0.22	.38	0.44	0.56	.31	.39	.44	0.94	1
1.5	B-251HP	3/8	1.63	5	6.31	0.5	0.31	.5	0.68	0.78	0.5	.66	.63	1.34	2
3	C-251HP	1/2	2	7.31	9.31	0.75	0.38	.75	0.94	1.19	0.75	.91	1	1.94	5
5	D-251HP	5/8	2.5	9.13	11.44	1	0.56	.88	1.13	1.56	1	1.28	1.19	2.5	10
8.5	EE-251HP	3/4	3	10.25	13.09	1.56	0.53	1.19	1.34	2.09	1.25	1.41	1.5	3.13	16
15	GG-251HP	1	4	15.25	19.13	1.75	0.81	1.5	1.75	3.5	2	2.03	2.13	4.91	43
35	HH-251HP	1 1/2	5	18.19	22.94	2	1.13	2	2.38	3.69	2.25	2.31	2.38	5.25	80
45	I-251HP		6	22.56	28.56	2.5	1.25	2.25	3	4	2.5	2.53	3	7	149
60	JJ-251HP		7	24	31	3	1.5	2.5	3.5	4.5	2.94	2.88	3.5	7	233
75	K-251HP		8	27	35	3.5	1.75	3	4	5	3.5	3.03	4	8.25	333
100	KK-251HP		10	36.5	47.5	4.06	2	4	5.5	6.75	4.5	4.03	5.5	11.38	731
200	L-251HP		11.25	35.93	48.31	5.06	3.09	4.5	6	8.25	5	4.81	6.38	13.5	885
300	M-251HP		14	44.2	59.2	6.13	3.94	4.75	7.5	9.5	6	6.06	8	17	1645

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

SWIVELS- HYDRO LINK\HP, HIGH PRESSURE, STAINLESS STEEL

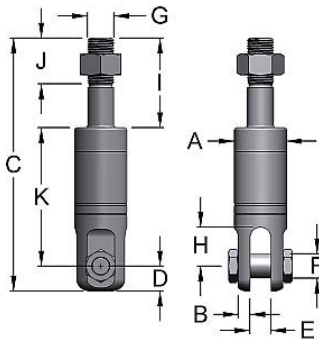
EYE TO EYE - TYPE 252HP, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-252HP	1/4	1.31	4.94	5.88	.44	0.31	.39	0.94	1
1.5	B-252HP	3/8	1.63	5	6.25	.69	0.5	.66	1.34	2
3	C-252HP	1/2	2	7.31	9.31	1	0.75	.91	1.94	5
5	D-252HP	5/8	2.5	9.06	11.44	1.19	1	1.28	2.5	10
8.5	EE-252HP	3/4	3	10.13	13.06	1.5	1.25	1.41	3.13	16
15	GG-252HP	1	4	15.13	19.38	2.13	2	2.03	4.91	43
35	HH-252HP	1 1/2	5	18.19	22.94	2.38	2.25	2.31	5.25	79
45	II-252HP		6	22.56	28.56	3	2.5	2.53	7	142
60	JJ-252HP		7	24	31	3.5	2.94	2.88	7	231
75	K-252HP		8	27	35	4	3.5	3.38	8.25	340
100	KK-252HP		10	36.5	47.5	5.5	4.5	4.03	11.38	704
200	L-252HP		11.25	35.56	48.31	6.38	5	4.81	13.5	875
300	M-252HP		14	43.2	59.2	8	6	6.06	17	1600

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

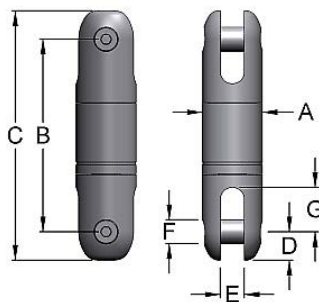
STUD TO CLEVIS - TYPE 351HP, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	H	I	J	K	Weight Lbs.
0.75	BB-351HP	1/4	1.31	0.22	5.81	.44	0.31	.38	0.63	0.56	2	1.5	4.38	1
1.5	B-351HP	3/8	1.63	0.31	7	.69	0.5	.5	0.75	0.78	2	1.5	4.31	2
3	C-351HP	1/2	2	0.38	10.44	.88	0.75	.75	0.88	1.19	3	2.63	5.25	5
5	D-351HP	5/8	2.5	0.56	11.19	1.13	1	.88	1.25	1.56	3.5	3.13	6.56	11
8.5	EE-351HP	3/4	3	0.53	13.78	1.34	1.56	1.19	1.5	2.09	4	3	8.44	15
15	GG-351HP	1	4	0.81	18.69	1.75	1.75	1.5	2	3.5	5.5	3	11.44	40
35	HH-351HP	1 1/2	5	1.13	25.56	2.38	2	2	2.5	3.69	8.5	4.25	14.69	84
45	II-351HP		6	1.25	28	3	2.5	2.25	3	4	8	4	17	141
60	JJ-351HP		7	1.5	31.5	3.5	3	2.5	3.25	4.5	8	4	20	219
75	K-351HP		8	1.75	36.5	4	3.5	3	3.75	5	10	5	22.5	295
100	KK-351HP		10	2	54.75	5.5	4.06	4	6.5	6.75	20.5	7.25	28.75	699

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

BULLET NOSE TO BULLET NOSE - TYPE 454HP, STAINLESS STEEL



WLL* Tons	Model Number	Rope Size	A	B	C	D	E	F	G	Weight Lbs.
0.75	BB-454HP	1/4	1.31	4.88	5.63	.44	0.31	.38	0.56	1
1.5	B-454HP	3/8	1.63	5	6.25	.56	0.5	.44	0.81	2
3	C-454HP	1/2	2	6.5	8.13	.81	0.75	.63	0.94	4
5	D-454HP	5/8	2.5	9.25	11.38	1.13	1	.88	1.56	9
8.5	EE-454HP	3/4	3	10.56	13.13	1.25	1.28	1	2.13	16
15	GG-454HP	1	4	15	18.5	1.75	1.75	1.5	3.25	42
35	HH-454HP	1 1/2	5	18.16	22.91	2.38	2	2	3.69	87

* Design factors for WLL=35T and below are 5:1; for 45T and above 4:1

MILLER CUSTOM SWIVELS



CUSTOM Y-LINK SWIVEL QUOTE REQUEST

Requestor

Name: _____ Date: _____ Quantity _____
 Company: _____ Address: _____
 City: _____ State: _____ Zip: _____
 Phone: _____ E-mail: _____

SWIVEL CONFIGURATION

CAP

CLEVIS__ EYE__ STUD__ BULLET__ SOCKET__ THIMBLE__ WEDGE__ HOOK__ EYEBOLT__

BARREL

STANDARD__ HEAVY DUTY__ HIGH PRESSURE__

SHANK

CLEVIS__ EYE__ STUD__ BULLET__ HOOK__ EYE BOLT__

SWIVEL MATERIAL

STANDARD__ STAINLESS STEEL__

SWIVEL CAPACITY & DIMENSIONS, WIRE ROPE

WORKING CAPACITY (LOAD LIMIT) _____
 (INDICATE US / METRIC TONS)

DESIGN FACTOR _____ WIRE ROPE SIZE _____

CLEVIS DIMENSIONS A _____ B _____ C _____

D _____

EYE DIMENSIONS E _____ F _____ G _____

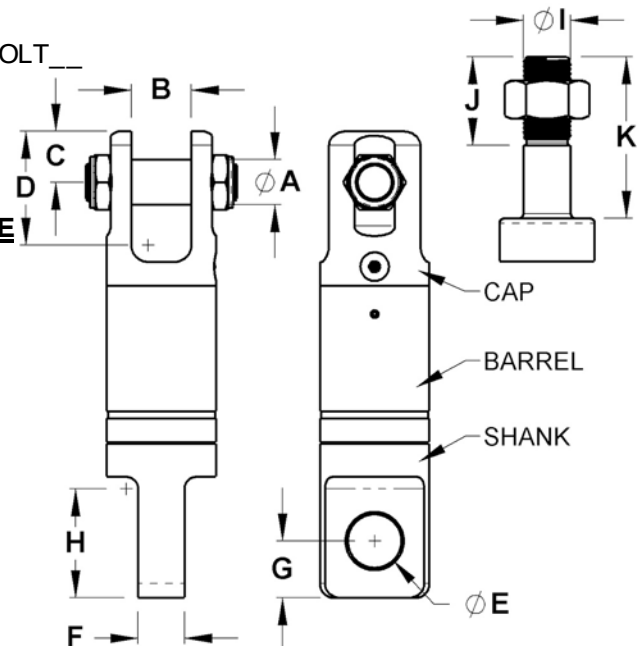
H _____

STUD DIMENSIONS I _____ J _____ K _____

OTHER REQUIEMENTS

ENVIRONMENT _____

SPECIAL TESTING, THIRD PARTY INSPECTION, ETC:



Miller Lifting Products

100A Sturbridge Rd.

Charlton, MA USA 01507

Tel.: (508) 248-3941 / Fax: (508) 248-0639

E-mail: sales@millerproducts.net

www.millerproducts.net

SWIVELS- PENGOMILLER, FOR LINE STRINGING



PENGO-MILLER Swivels



90 degree connectors

- The original PENGOMILLER tension line stringing swivels and 90° connectors.
- Designed specifically for horizontal pulling of tension lines and passage over sheaves, overhead or through conduit.
- Swivels have Working Loads up to 100,000 lbs. with a 3:1 design factor.
- Equipped with low friction angular contact ball bearings.
- Machined, heat treated from solid forged alloy steel bar. Single-piece shank.
- Zinc plating inside and out for effective protection against corrosion.
- Hex socket pins are standard. Optional slotted pins available for sizes A thru D.
- Factory lubricated and sealed with neoprene seals.
- 90° Connectors permit limited movement of connected cables in two perpendicular planes. Short, bullet shape accommodates passage through blocks. Mean Breaking Load up to 150,000 lbs.
- Order spare pins by type (hex or slotted), and by model number of the swivel or connector. For example, "two hex pins for A-13-L" or "two slotted pins for B-13-L-SL".



Green Dot = Proof Load Tested

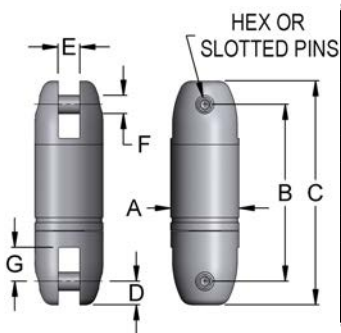
NEW

PROOF LOAD TESTED

All swivel models A-13-L, BB-13-L and B-13-L are individually proof load tested to 120% of their working load limit as indicated by a green proof test marking. Test certificates available. Other sizes tested upon request.

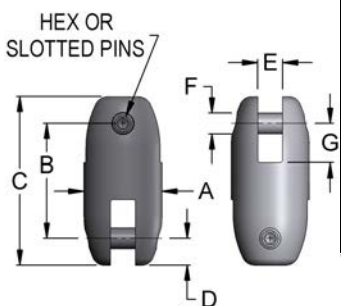
Safety Note: These swivels are designed for straight line pulling and will not support side loads as would be encountered traveling over a bull wheel. Miller line stringing swivels are not intended for general lifting applications. Swivel selection should be based on pulling line diameter, grip size, opening dimensions, pin sizes, working load, and sheave groove diameter. For general lifting applications, see Miller Y-Link and Miller Econo-Link Swivels.

LINE STRINGING SWIVELS

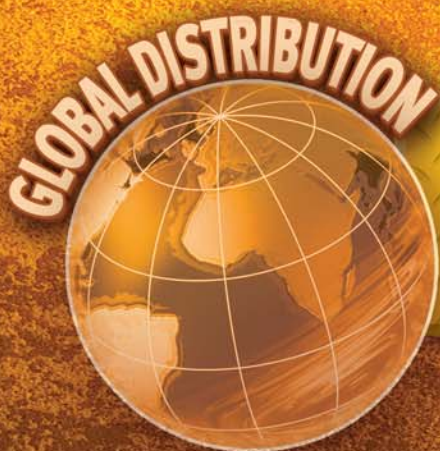


Model # Hex Pin	Model # Slotted Pin	WLL* Lbs.	A	B	C	D	E	F	G	Weight Lbs.
A13L	A13L-SL	1,800	.875	2.34	3.09	0.375	0.31	.31	0.47	0.38
BB13L	BB13L-SL	4,000	1.25	3.66	4.51	0.44	0.59	.38	0.69	1
B13L	B13L-SL	7,500	1.44	4	5.13	0.56	0.59	.44	0.81	1.38
C13L	C13L-SL	10,000	1.875	5.44	7.06	0.81	0.75	.62	0.94	3.63
D13L	D13L-SL	16,000	2.44	7.88	10.19	1.125	1	.88	1.56	8
EE13L	N/A	25,000	2.875	9.81	12.31	1.25	1.28	1	2.13	15
D13-4BL	N/A	30,000	2.5	8.91	11.16	1.125	1	.88	1.56	9
GG13L	N/A	50,000	3.875	13.25	16.75	1.75	1.75	1.5	3.5	40
HH13L	N/A	100,000	4.875	15.81	20.56	2.375	2	2	3.69	78

90° CONNECTORS



Model # Hex Pin	Model # Slotted Pin	MBL Lbs.	ROPE	A	B	C	D	E	F	G	Weight Lbs.
PL 1/4	PL 1/4-SL	5,400	1/4	.88	1.31	2.06	.38	0.31	.31	0.47	0.19
PL 3/8	PL 3/8-SL	12,000	3/8	1.25	1.75	2.63	.44	0.59	.38	0.69	0.38
PL 1/2	PL 1/2-SL	22,500	1/2	1.44	2.13	3.25	.56	0.59	.44	0.81	0.69
PL 5/8	PL 5/8-SL	30,000	5/8	1.88	2.5	4.13	.81	0.75	.62	0.94	1.69
PL 3/4	PL 3/4-SL	60,000	3/4	2.44	3.38	5.63	1.13	1	.88	1.31	3.75
PL 7/8	N/A	75,000	7/8	2.63	3.88	6.38	1.25	1	1	1.5	6.5
PL 1	N/A	150,000	1	3.88	8	11.5	1.75	1.75	1.5	3.5	18.5



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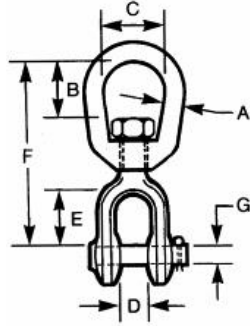
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10th Edition

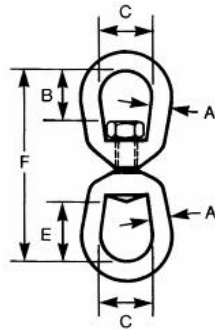
DROP FORGED SWIVELS



JAW & EYE SWIVELS

Size (A) in Inches	Working Load Limit in Pounds	Approximate Weight Each in Pounds	Dimensions in Inches					
			B	C	D	E	F	G
1/4	850	0.22	0.69	0.75	0.38	0.88	2.69	0.25
5/16	1,250	0.39	0.88	1.00	0.50	0.88	2.88	0.31
3/8	2,250	0.71	0.94	1.25	0.63	1.00	3.50	0.38
1/2	3,600	1.40	1.38	1.50	0.81	1.31	4.50	0.50
5/8	5,200	2.30	1.63	1.75	1.00	1.50	5.31	0.63
3/4	7,200	3.50	1.75	2.00	1.19	1.75	6.06	0.75
7/8	10,000	5.70	2.08	2.25	1.20	2.07	7.06	0.88
1	12,500	9.50	2.27	2.44	1.73	2.85	8.56	1.12
1.1/4	18,000	16.00	2.69	3.13	2.06	2.81	9.44	1.38
1.1/2	45,200	55.00	4.20	4.00	2.87	4.43	14.75	2.25

Meets or exceeds the performance requirements of Federal Specification RR-C-271, current revision. Hot galvanized.



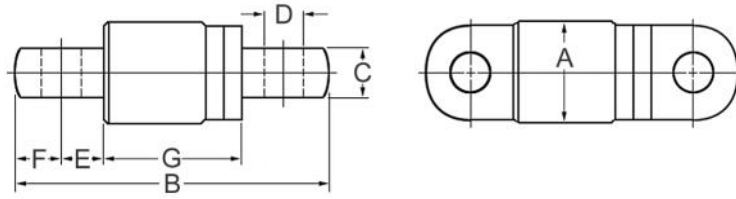
EYE & EYE SWIVELS

Size (A) in Inches	Working Load Limit in Pounds	Approximate Weight Each in Pounds	Dimensions in Inches			
			B	C	E	F
1/4	850	0.22	0.69	0.75	0.94	2.88
5/16	1,250	0.39	0.75	1.00	1.13	3.63
3/8	2,250	0.71	0.94	1.25	1.38	4.25
1/2	3,600	1.40	1.38	1.50	1.94	5.63
5/8	5,200	2.30	1.63	1.75	2.31	6.63
3/4	7,200	3.50	1.81	2.00	2.56	7.25
7/8	10,000	6.00	2.06	2.25	3.02	8.28
1	12,500	8.50	2.35	2.48	3.43	9.53
1.1/4	18,000	16.50	2.69	3.13	3.69	11.13
1.1/2	45,200	46.00	4.18	4.00	4.18	17.12

Meets or exceeds the performance requirements of Federal Specification RR-C-271, current revision. Hot galvanized.



COMPACT THRUST BEARING EYE & EYE SWIVEL



Size	A	B	C	D	E	F	G	Working Load Limit	Weight in Pounds
3 Ton	2.13	7.48	1.06	1.02	1.06	1.10	3.15	6,000	5.5
5 Ton	2.32	8.66	1.30	1.18	1.22	1.30	3.62	10,000	6.6
8 Ton	3.35	10.43	1.65	1.34	1.34	1.54	4.65	16,000	15.7
12 Ton	3.70	11.61	1.97	1.61	1.61	1.77	4.84	24,000	22.5
17 Ton	4.69	14.53	2.05	2.05	2.28	2.17	5.63	34,000	39.7
22 Ton	4.92	16.97	2.28	2.05	2.17	2.48	7.68	44,000	59.0

